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Kim et al.

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(54) **MOBILE TERMINAL**

(56) **References Cited**

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U.S. PATENT DOCUMENTS

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11,016,532 B2 * 5/2021 Yang H04M 1/0268
11,720,141 B2 * 8/2023 Kim H04N 23/54
345/173

(Continued)

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FOREIGN PATENT DOCUMENTS

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KR 10-2013-0099449 9/2013
KR 10-2016-0081786 7/2016

(Continued)

OTHER PUBLICATIONS

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§ 371 (c)(1),

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(57) **ABSTRACT**

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H04B 1/38 (2015.01)
H04M 1/02 (2006.01)

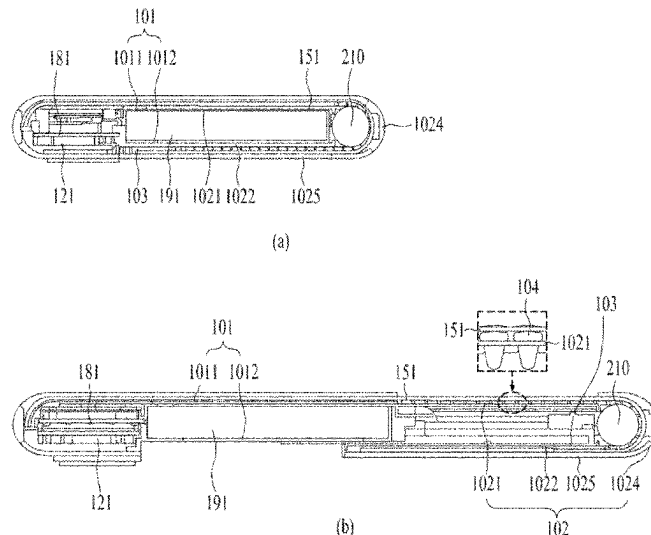
(52) **U.S. Cl.**
CPC **H04M 1/0237** (2013.01); **H04M 1/0268** (2013.01); **H04M 2201/38** (2013.01)

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See application file for complete search history.

Provided is a mobile terminal switchable from a first state to a second state extended in a first direction, the mobile terminal comprising: a first frame; a second frame slidable in the first direction with respect to the first frame; a roller coupled to the end of the second frame in the first direction; a flexible display including a fixing part coupled to the first frame and a variable part extends from the fixing part and surrounding the roller, and is positioned on the front surface according to the sliding movement of the second frame; and a rolling hinge positioned on the rear surface of the variable part and changing correspondingly to the shape of the variable part, wherein the roller includes a sawtooth part formed on the outer surface thereof, and the rolling hinge includes a concave-convex part corresponding to the sawtooth part.

9 Claims, 21 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2005/0225093	A1 *	10/2005	Zuo	H04M 1/0216
				292/1
2009/0133216	A1 *	5/2009	Cho	E05D 11/1064
				16/250
2014/0211399	A1 *	7/2014	O'Brien	G06F 1/1652
				29/592.1
2019/0268455	A1 *	8/2019	Baek	G06F 1/1652

FOREIGN PATENT DOCUMENTS

KR	10-2019-0062855	6/2019
KR	10-2019-0077107	7/2019
KR	10-2019-0101605	9/2019
KR	10-2020-0099455	8/2020

* cited by examiner

FIG. 1

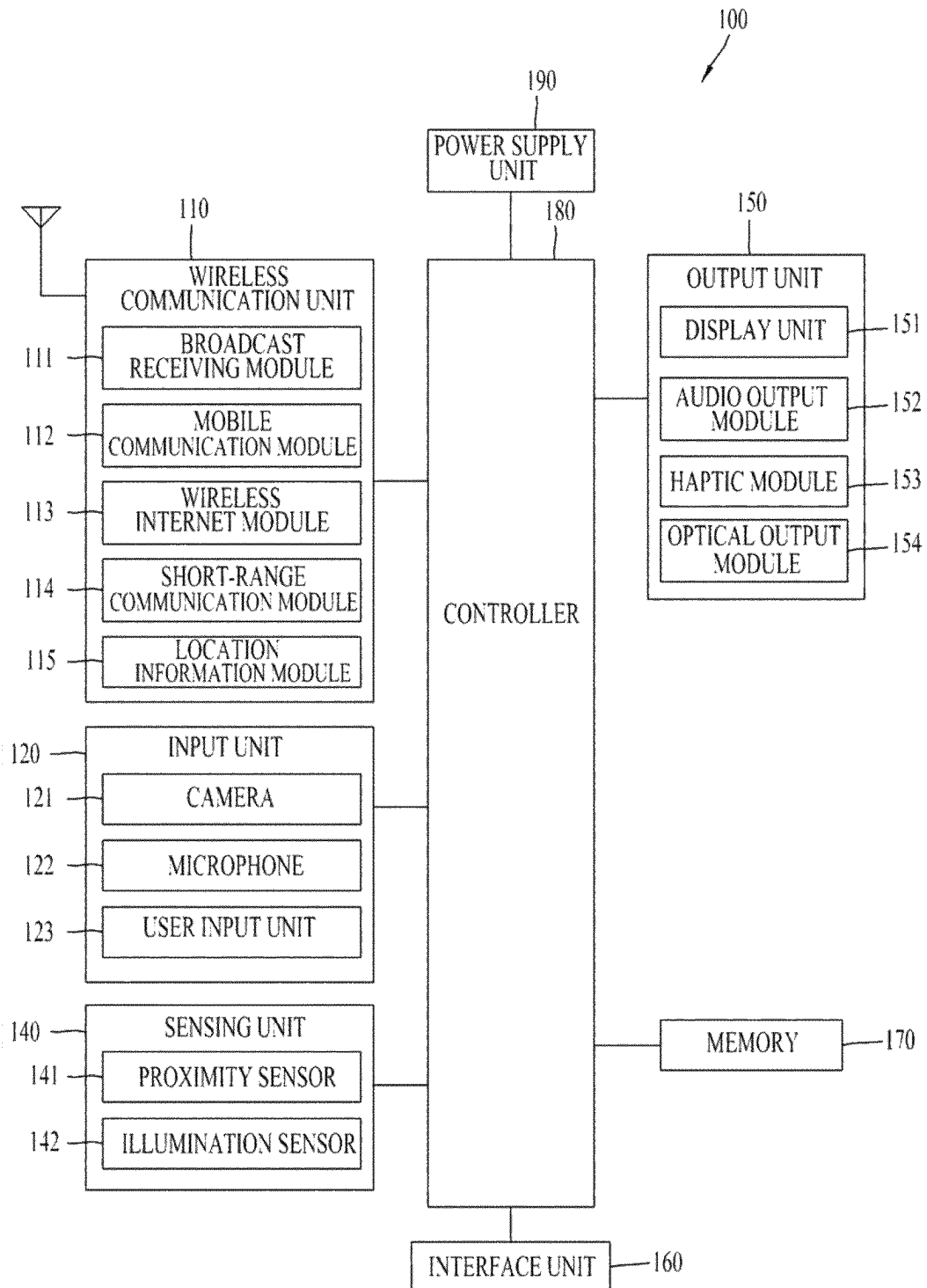


FIG. 2

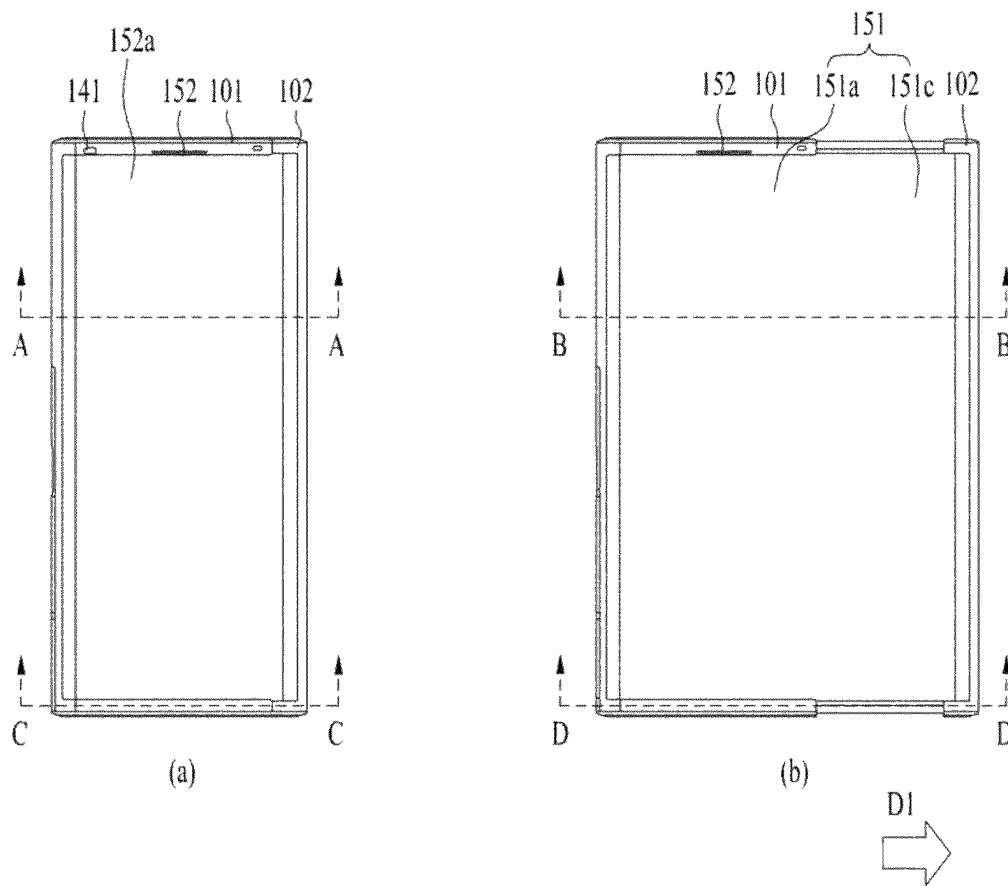
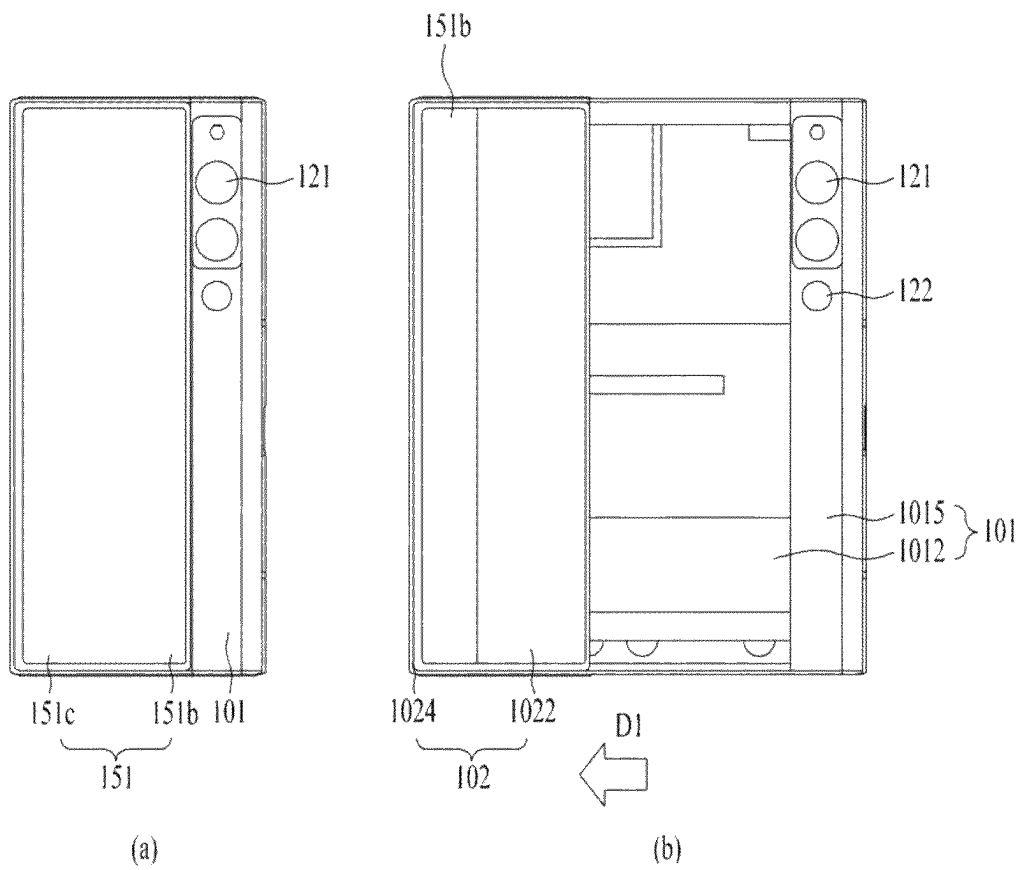


FIG. 3



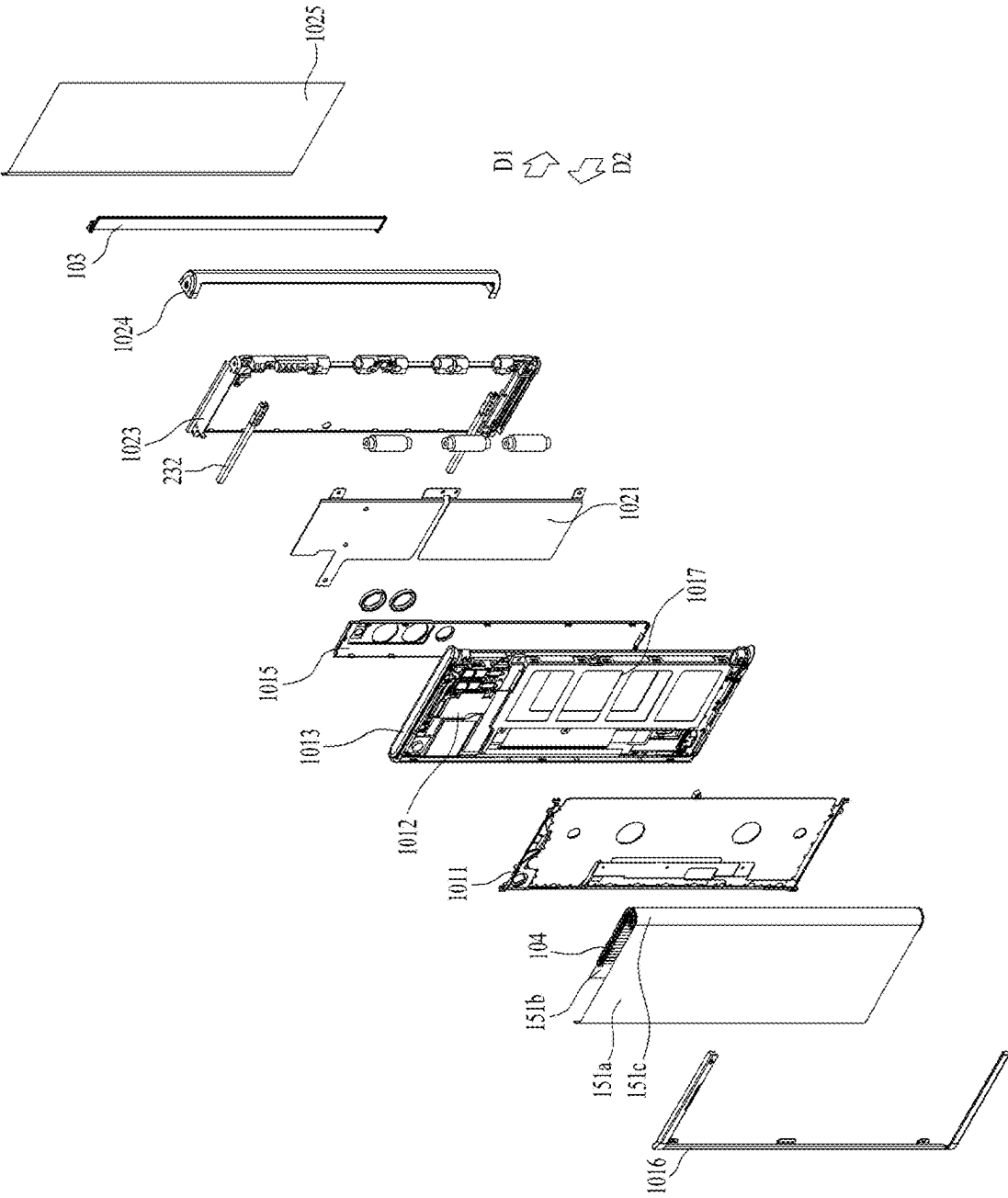


FIG. 4

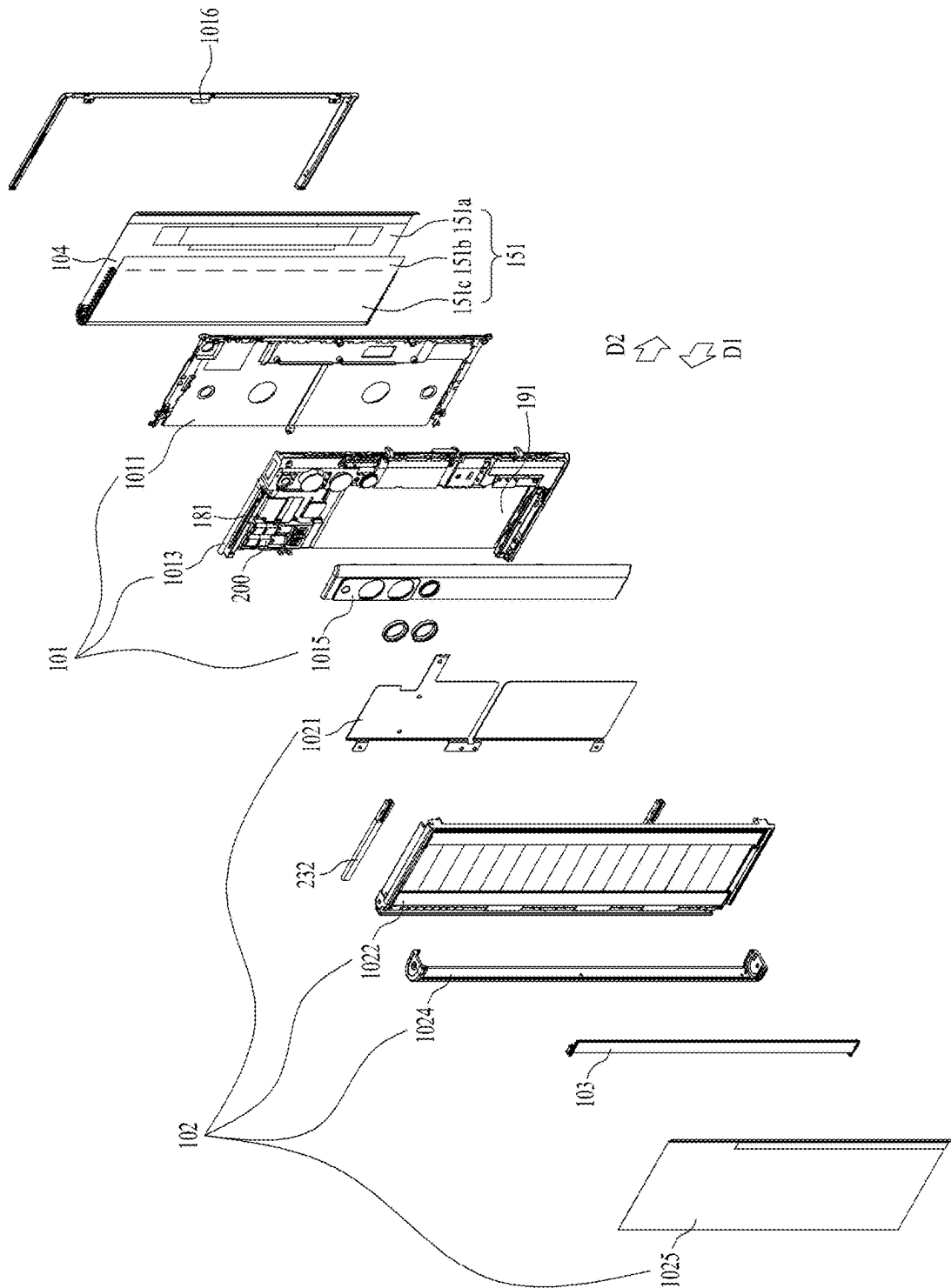


FIG. 5

FIG. 6

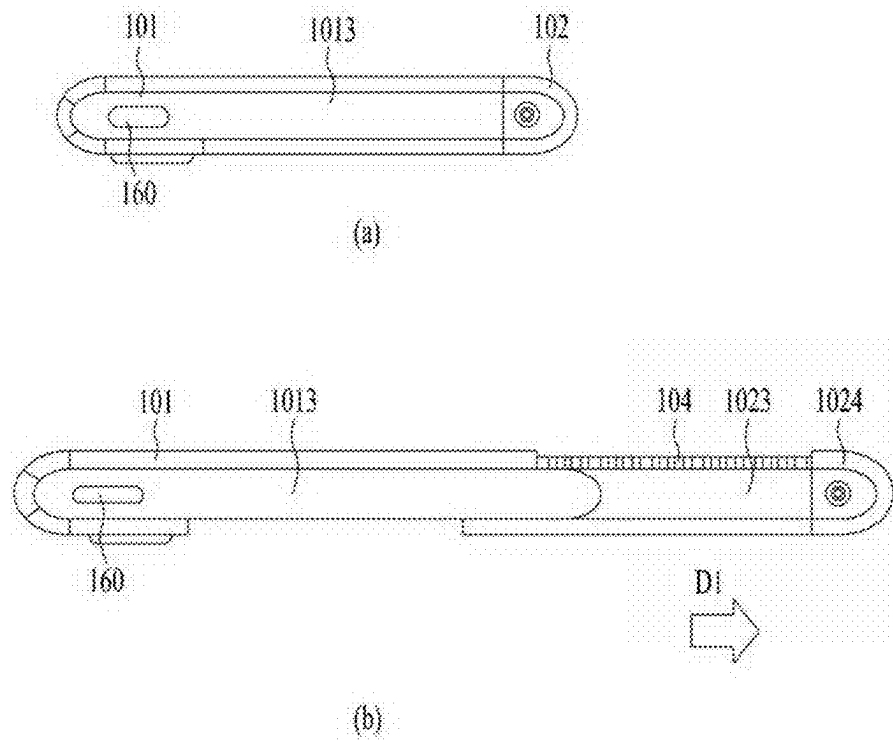


FIG. 7

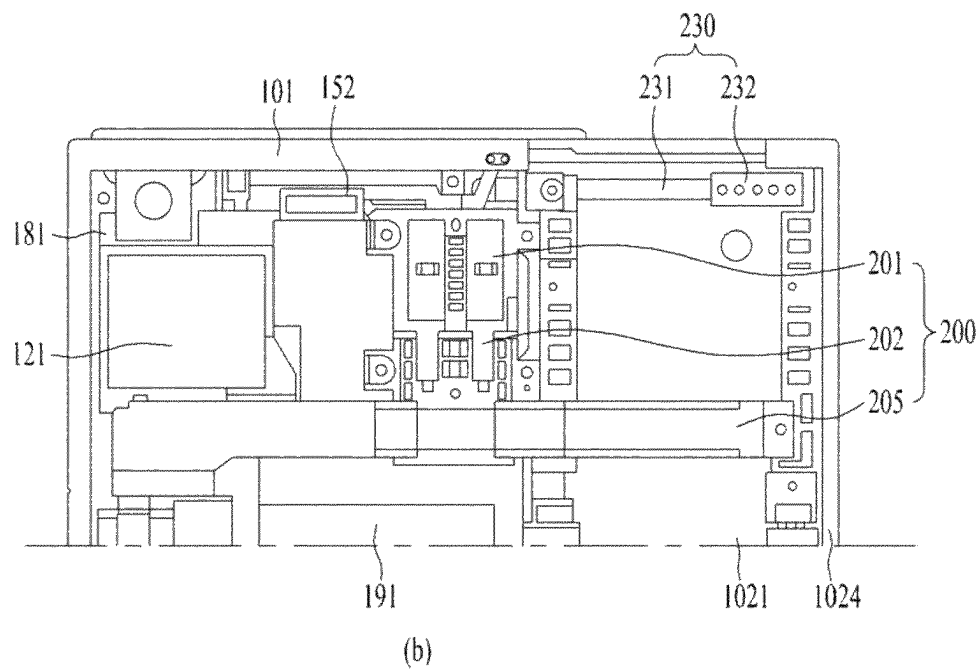
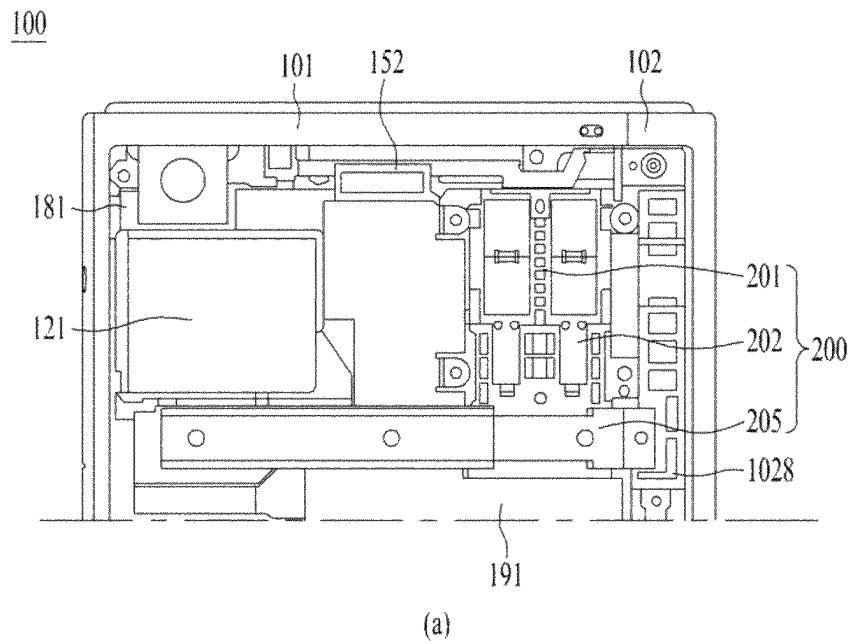
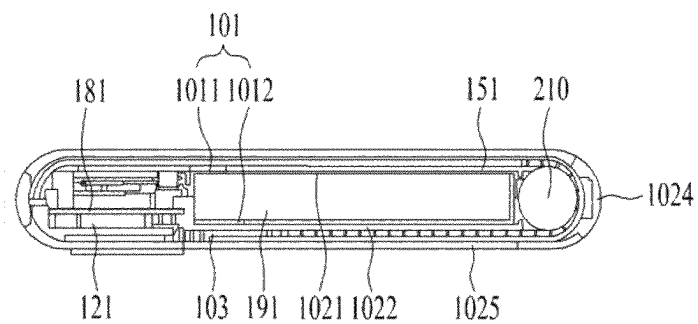
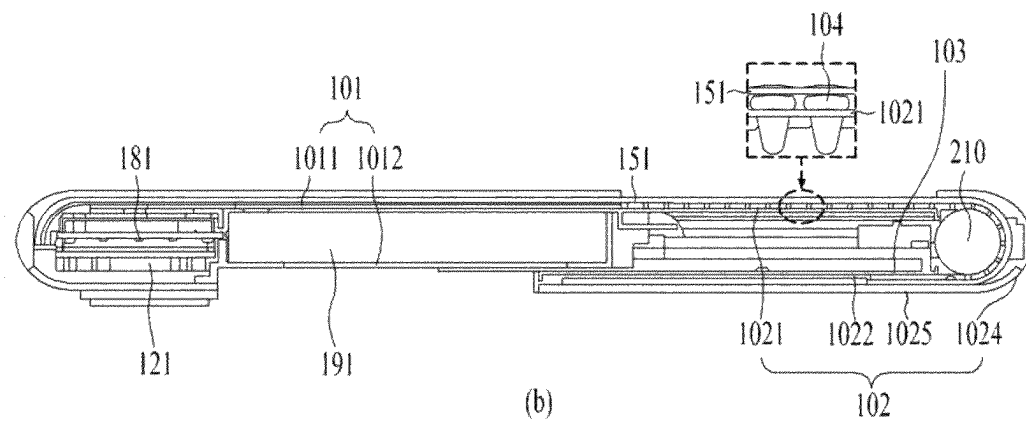


FIG. 8



(a)



(b)

FIG. 9

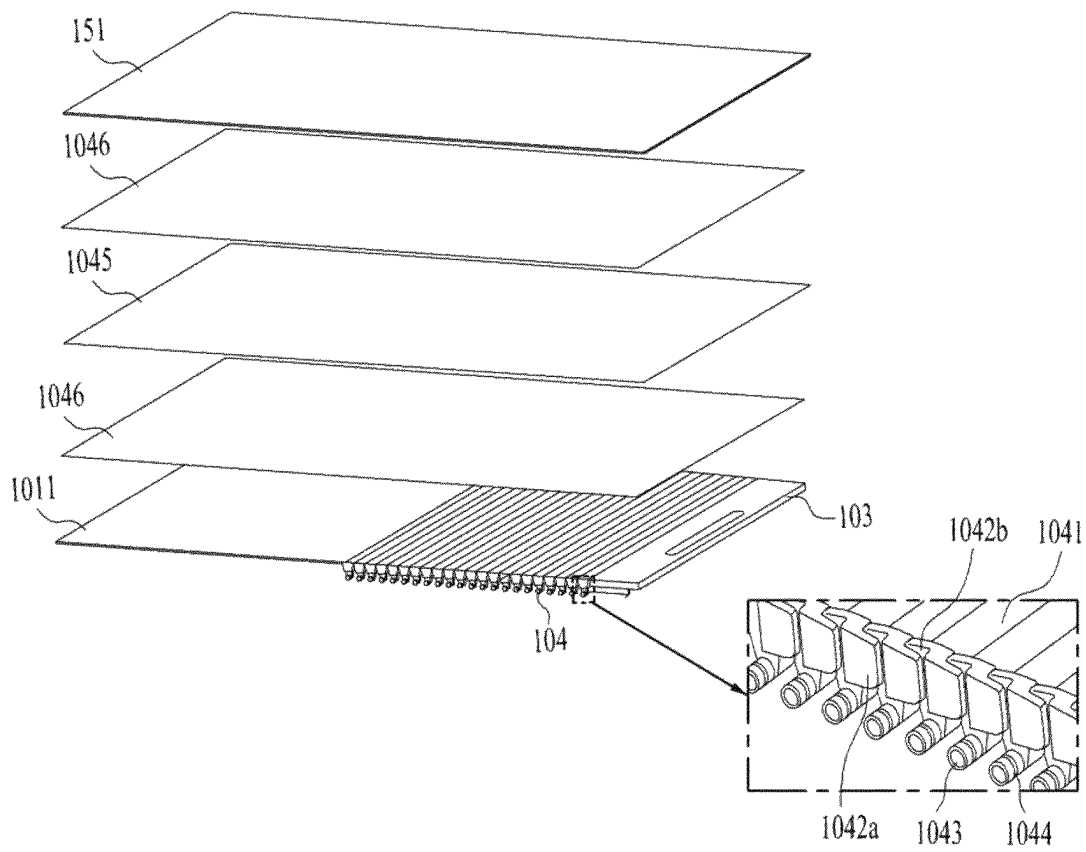
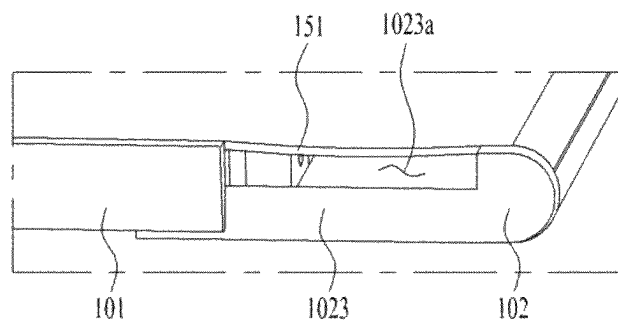
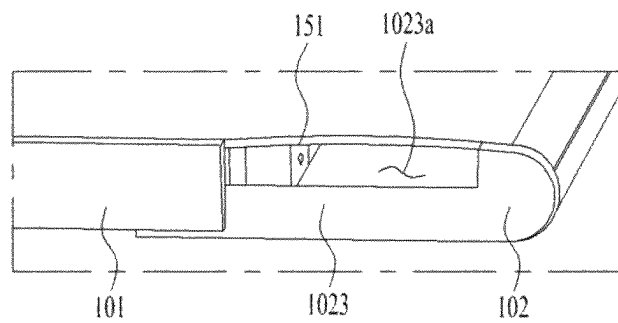


FIG. 10



(a)



(b)

FIG. 11

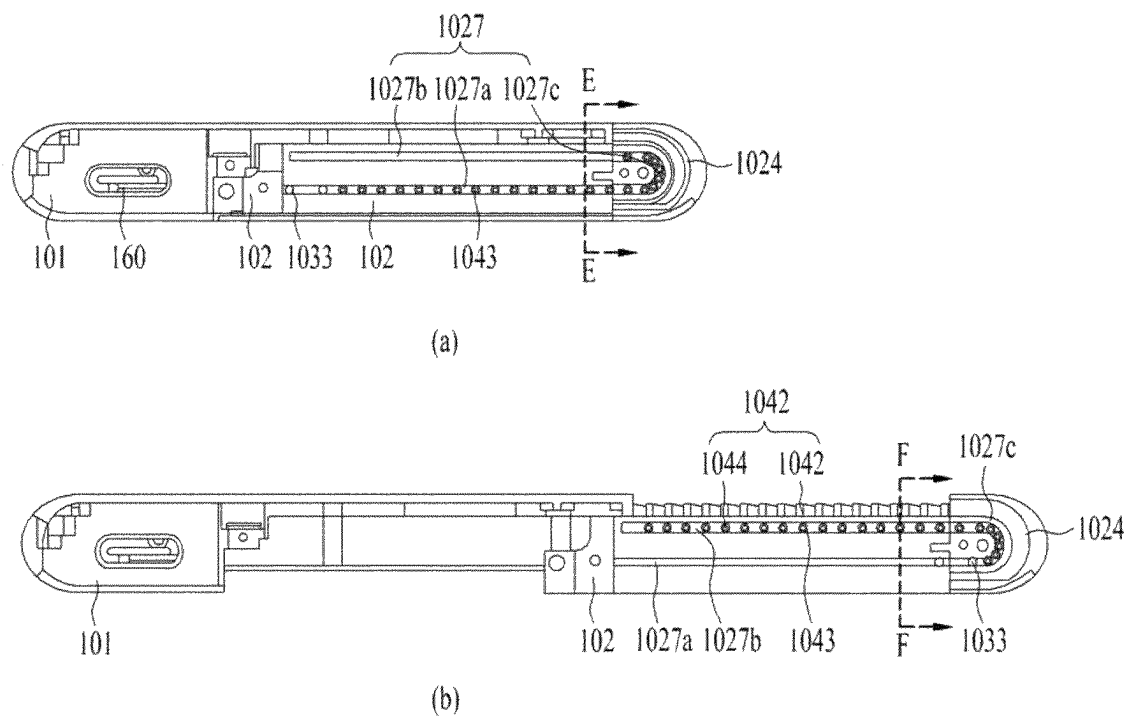
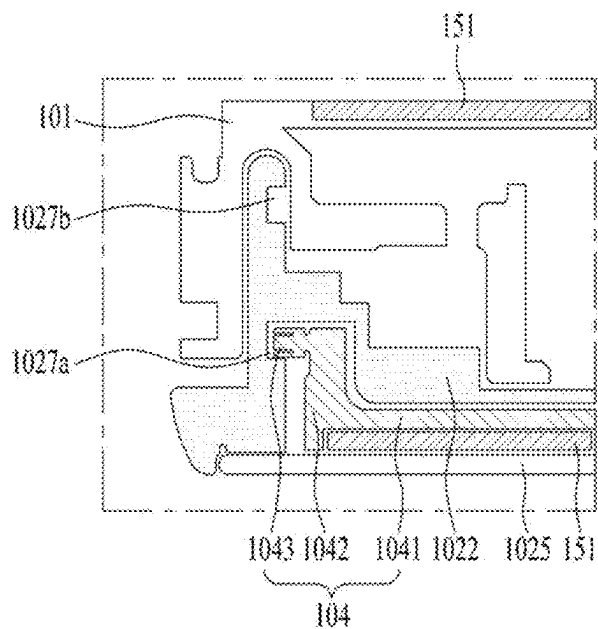
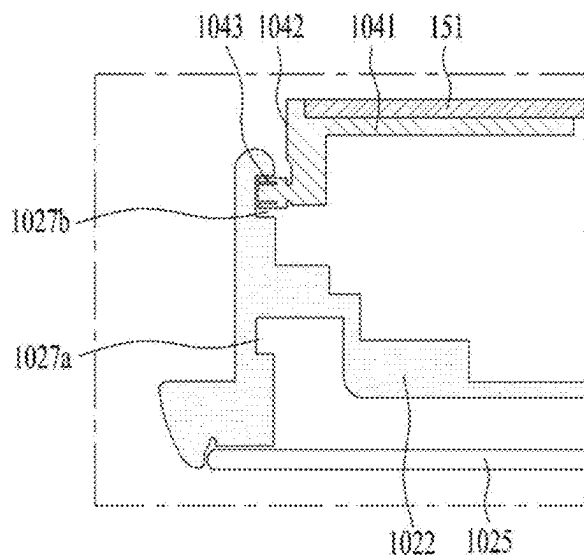


FIG. 12



(a)



(b)

FIG. 13

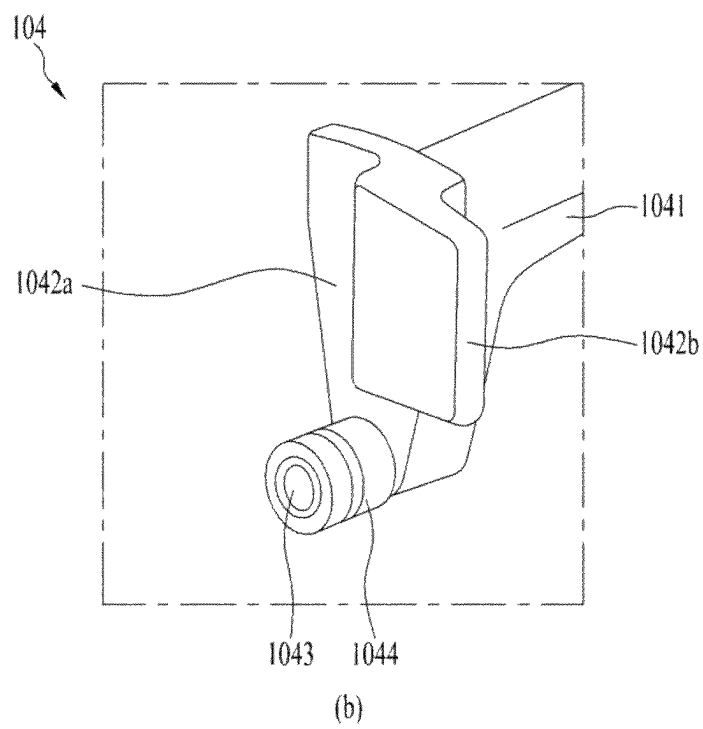
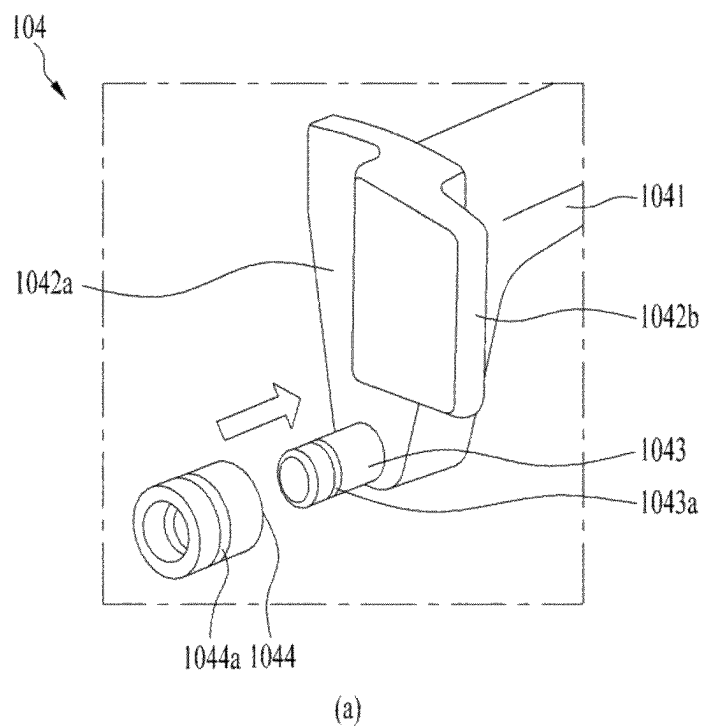


FIG. 14

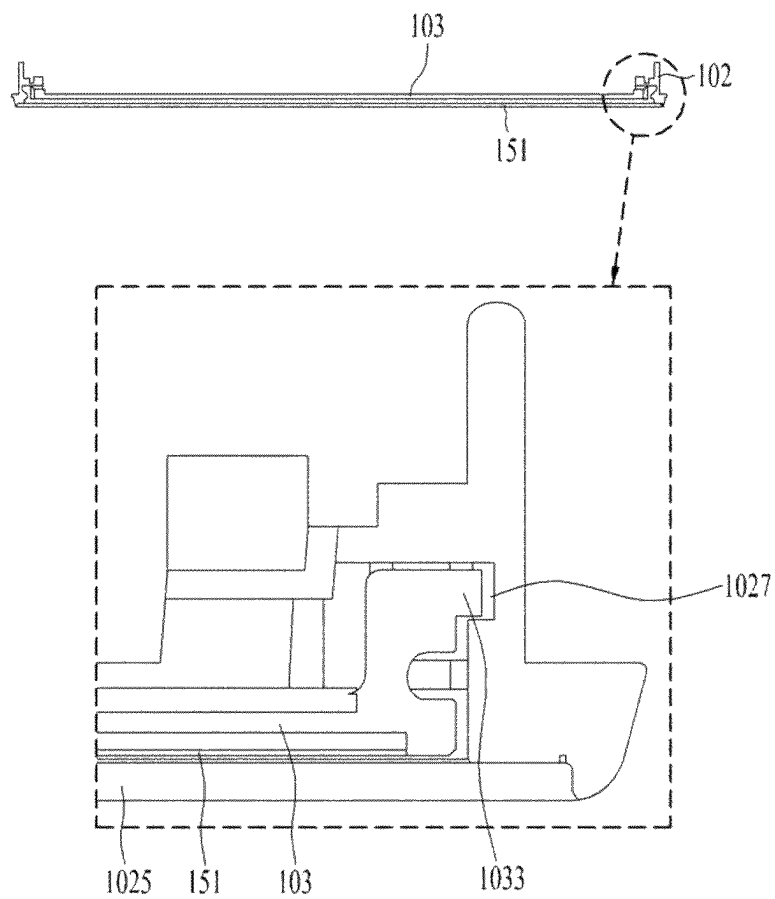


FIG. 15

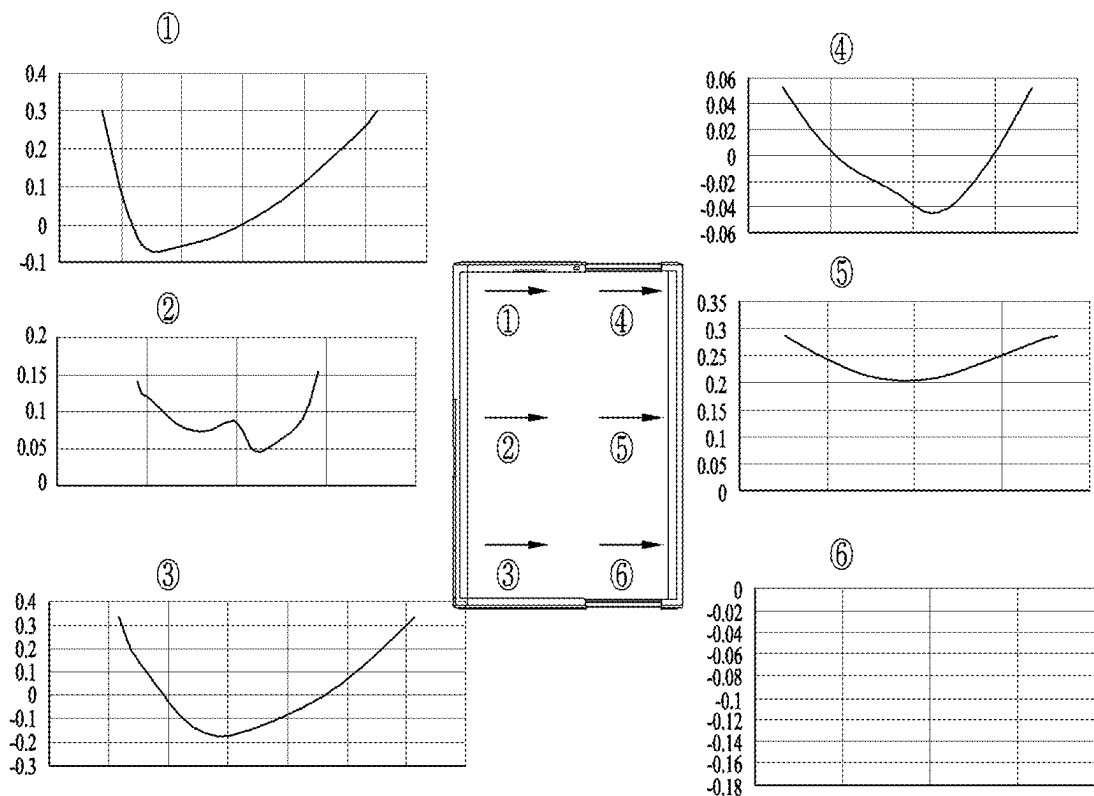


FIG. 16

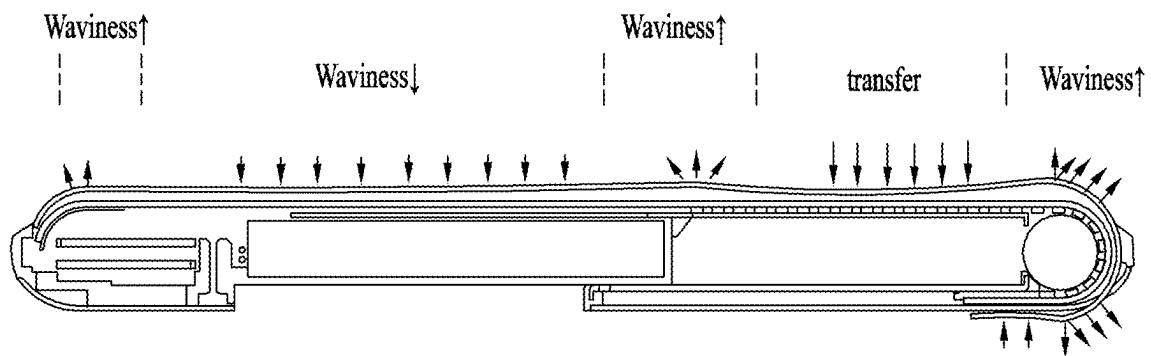


FIG. 17

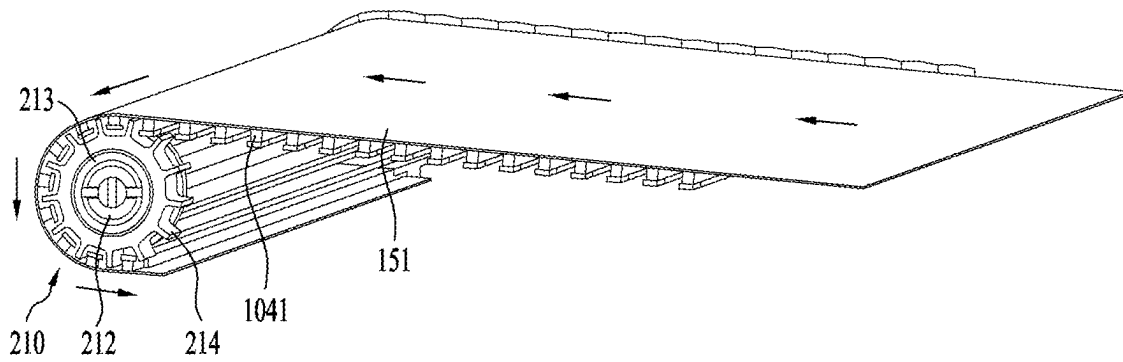


FIG. 18

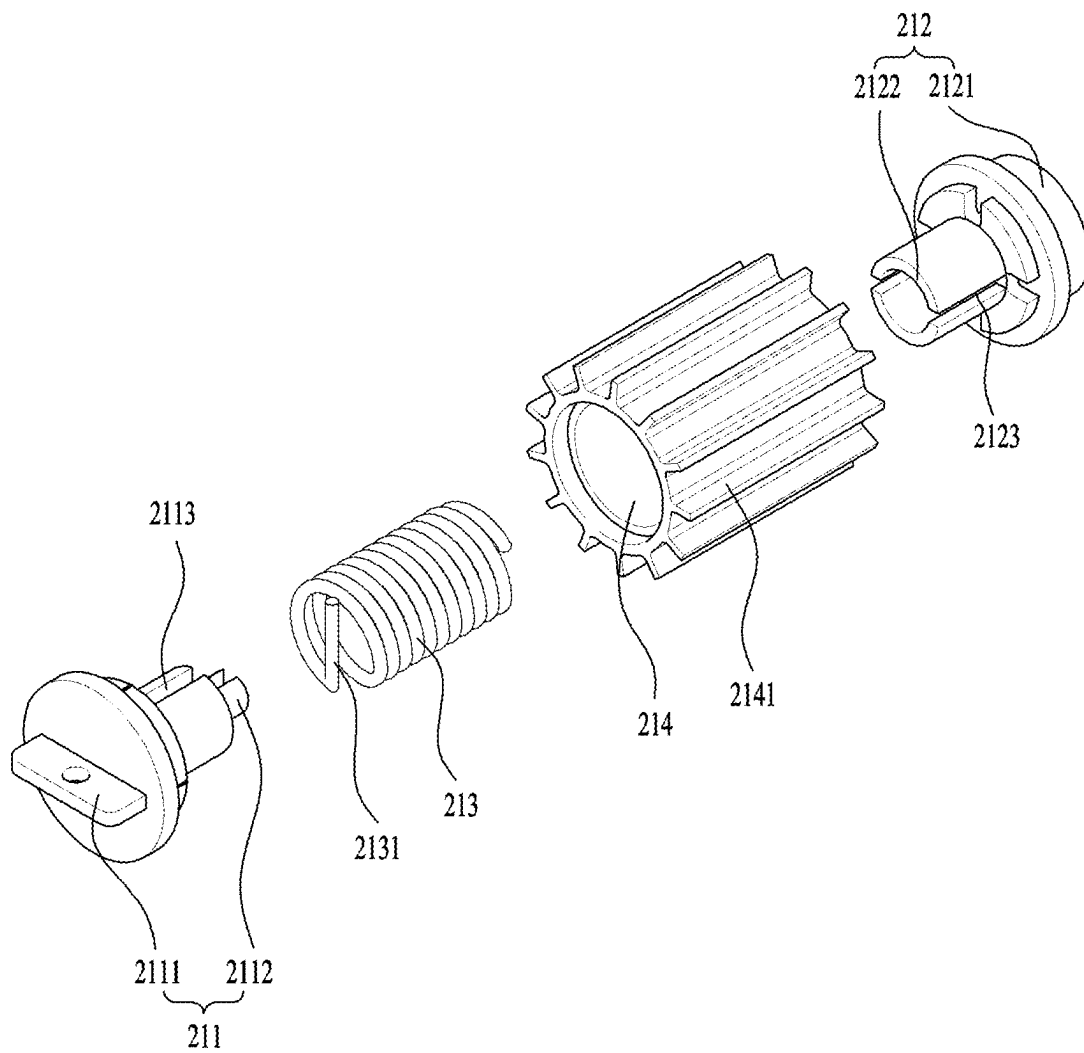


FIG. 19

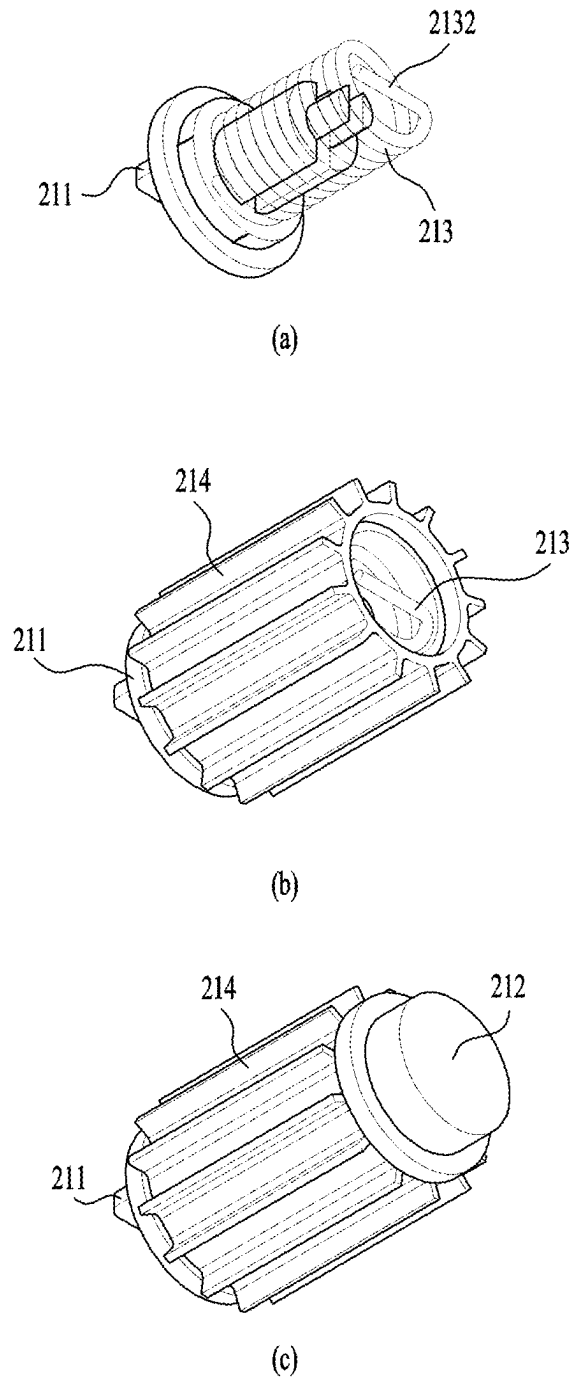
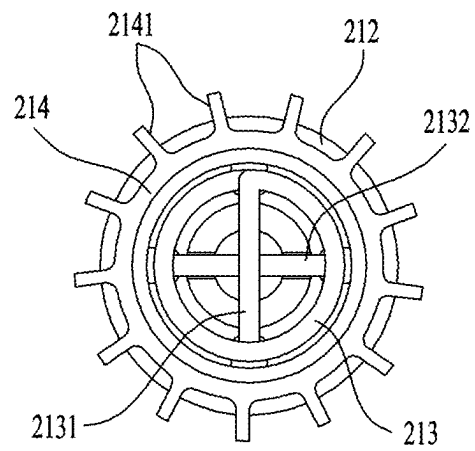
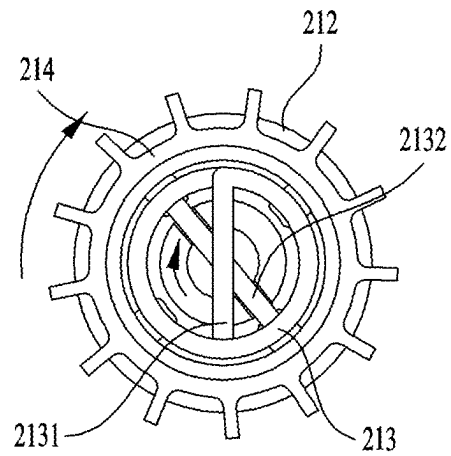


FIG. 20

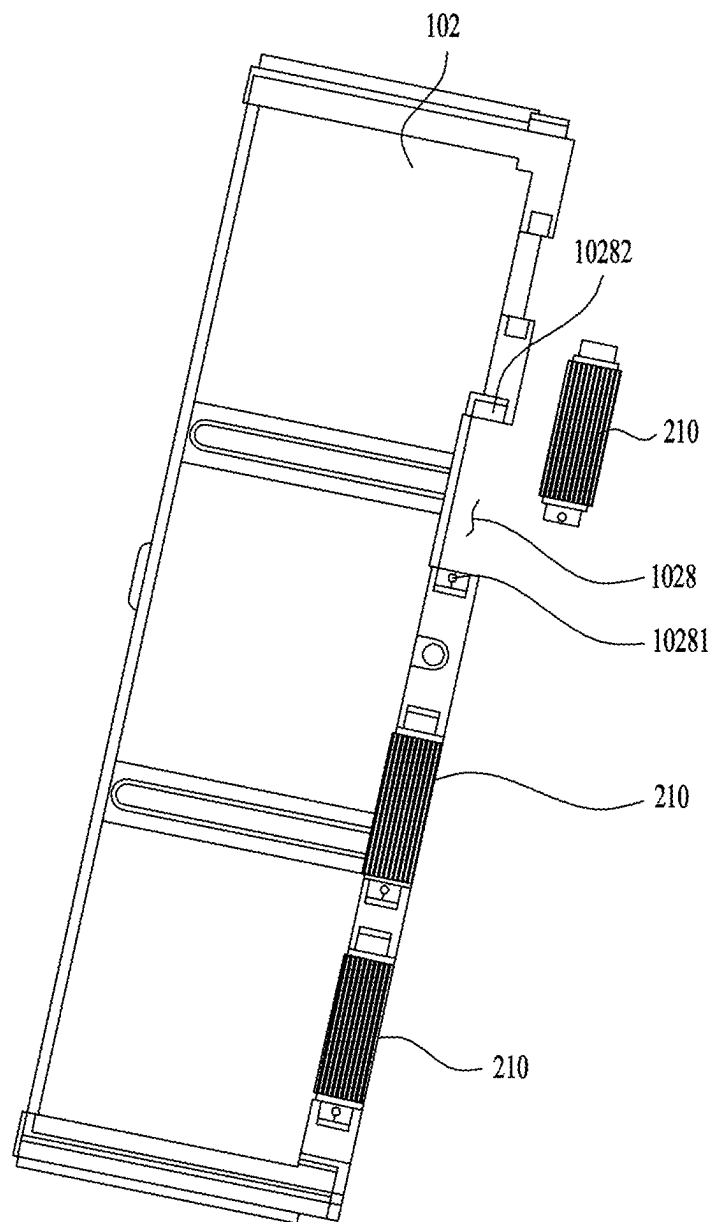


(a)



(b)

FIG. 21



1

MOBILE TERMINAL

CROSS-REFERENCE TO RELATED
APPLICATIONS

This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2020/012362, filed on Sep. 14, 2020, the contents of which are all incorporated by reference herein in their entirety.

TECHNICAL FIELD

The present disclosure relates to a mobile terminal, and more particularly, to a mobile terminal that has a flexible display and is capable of extending a size of a screen while the display is scrolled and slid at the same time.

BACKGROUND

Terminals may be generally classified as mobile/portable terminals or stationary terminals according to their mobility. Mobile terminals may also be classified as handheld terminals or vehicle mounted terminals according to whether or not a user can directly carry the terminal.

A display device is a device having a function of receiving, processing, and displaying a video that a user may watch. For example, the display device receives a broadcast selected by the user from broadcast signals transmitted from a broadcasting station, separates a video signal from the received signals, and displays the separated video signal on a display.

In recent years, because of a development of a broadcasting technology and a network technology, functions of the display device have also been considerably diversified, and a performance of the device has been improved accordingly. That is, the display device has been developed to provide not only broadcast contents but also various other contents to the user. For example, the display device may provide game play, music listening, internet shopping, user customized information, and the like using various applications as well as programs received from the broadcasting station. In order to perform such extended functions, the display device may be basically connected to other devices or networks using various communication protocols, and may provide the user with a ubiquitous computing environment. In other words, the display device has evolved into a smart device that enables connectivity to a network and continuous computing.

Recently, a flexible display having sufficient elasticity and capable of large deformation has been developed. The size of a mobile terminal can be varied using the deformable nature of the flexible display. For the mobile terminal having such a variable structure, changing the size of the mobile terminal should be stably performed, and there is a need for a structure to support the extended display unit to maintain a flat state of the display unit.

SUMMARY

Technical Problem

An object of the present disclosure is to provide a mobile terminal for improving the durability of a flexible display by preventing a point at which the flexible display is bent from being limited to a specific position.

2

Another object of the present disclosure is to improve the utility of a flexible display by preventing the flexible display from being wavy.

Technical Solutions

Provided is a mobile terminal capable of switching a state thereof from a first state to a second state extended in a first direction including a first frame, a second frame capable of sliding in a first direction or in a second direction opposite to the first direction relative to the first frame, a roller coupled to an end in the first direction of the second frame, a flexible display unit including a fixed portion coupled to the first frame and a variable portion extending in a rearward direction from the fixed portion while surrounding the roller and having an area located at the front variable based on the sliding movement of the second frame, and a rolling hinge located on a rear surface of the variable portion and changing based on a shape of the variable portion, wherein the roller includes teeth formed on an outer surface thereof, wherein the rolling hinge includes irregularities corresponding to the teeth.

The rolling hinge may include a plurality of support bars extending in a third direction perpendicular to the first direction and arranged adjacent to each other in the first direction, and the teeth of the roller may include each protrusion inserted between adjacent two of the plurality of support bars.

The roller may include a fixed shaft fixed to the second frame, a driving shaft rotating relative to the fixed shaft, a coil spring having one side coupled to the fixed shaft and the other side coupled to the driving shaft, and a rotating cylinder rotating together with the driving shaft and having the teeth formed thereon.

The rotating cylinder and the driving shaft may rotate when the second frame moves and change the number of turns of the coil spring, and the coil spring may provide elasticity to restore the reference number of turns to the driving shaft and the rotating cylinder.

The elasticity of the coil spring may be provided when restoring the first state from the second state.

The second frame may include a roller seating portion for accommodating the roller therein at the end in the first direction thereof, and the roller seating portion may include a fixing hole for receiving a screw fastened through one end of the fixed shaft, and a bearing for one end of the driving shaft to be rotatably seated.

At least a portion of the driving shaft may rotatably overlap the fixed shaft inside of the coil spring.

The fixed shaft may include a first coupling groove for receiving one end of the coil spring, and the driving shaft may include a second coupling groove for receiving the other end of the coil spring.

Both of the ends of the coil spring may be bent in a radial direction of the coil, and the first coupling groove and the second coupling groove may extend through a center of rotation of the driving shaft.

The first coupling groove and the second coupling groove may respectively extend to the other end of the fixed shaft and the other end of the driving shaft.

The roller may include a plurality of rollers spaced apart from each other in a third direction perpendicular to the first direction.

Advantageous Effects

The mobile terminal according to the present disclosure may adjust the size of a screen according to the need, thereby satisfying both portability and utility.

The mobile terminal according to the present disclosure may prevent the display from being damaged since stress is not concentrated at a specific point of the display.

The mobile terminal may prevent the display from being wavy when the second frame moves, that is, keep the flatness of the display when the second frame moves.

The mobile terminal includes a structure for guiding the frames to move in parallel, thereby achieving stable state transition. Further scope of applicability of the present disclosure will become apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the disclosure, are given by illustration only, since various changes and modifications within the spirit and scope of the disclosure will become apparent to those skilled in the art from this detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of a mobile terminal in accordance with the present disclosure;

FIG. 2 is a front view of a first state and a second state of the mobile terminal in accordance with an embodiment;

FIG. 3 is a rear view of the first state and the second state of the mobile terminal in accordance with an embodiment;

FIGS. 4 and 5 are exploded perspective views of the mobile terminal in accordance with an embodiment;

FIG. 6 is a side view of the mobile terminal as viewed from a third direction;

FIG. 7 is a view showing a driving unit of the mobile terminal in accordance with an embodiment;

FIG. 8 is a cross-sectional view taken along lines A-A and B-B in FIG. 2;

FIG. 9 is a view illustrating a display unit and a rolling plate of the mobile terminal in accordance with an embodiment;

FIG. 10 is a view illustrating an issue raised in an extendable mobile terminal;

FIG. 11 is a cross-sectional view taken along lines C-C and D-D in FIG. 2;

FIG. 12 is a cross-sectional view taken along lines E-E and F-F in FIG. 11;

FIG. 13 is a view showing an expanded side portion and a guide hook of the mobile terminal in accordance with an embodiment;

FIG. 14 is a view showing a rolling hinge of the mobile terminal with an embodiment;

FIG. 15 is a graph showing phase distortion in the mobile terminal;

FIG. 16 is a conceptual view for explaining causes of the phase distortion of FIG. 15;

FIG. 17 is a view showing coupling of a roller and a rolling hinge of a body of a mobile terminal;

FIG. 18 is an exploded perspective view of a roller of a mobile terminal;

FIG. 19 is a view showing an order of fastening each component of a roller;

FIG. 20 is a view showing a cross-section based on rotation of a roller; and

FIG. 21 is a view showing arrangement of rollers coupled to a second frame of the present disclosure.

DETAILED DESCRIPTION

Description will now be given in detail according to exemplary embodiments disclosed herein, with reference to

the accompanying drawings. For the sake of brief description with reference to the drawings, the same or equivalent components may be provided with the same reference numbers, and description thereof will not be repeated. In general, a suffix such as “module” and “unit” may be used to refer to elements or components. Use of such a suffix herein is merely intended to facilitate description of the specification, and the suffix itself is not intended to give any special meaning or function. In the present disclosure, that which is well-known to one of ordinary skill in the relevant art has generally been omitted for the sake of brevity. The accompanying drawings are used to help easily understand various technical features and it should be understood that the embodiments presented herein are not limited by the accompanying drawings. As such, the present disclosure should be construed to extend to any alterations, equivalents and substitutes in addition to those which are particularly set out in the accompanying drawings.

It will be understood that although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are generally only used to distinguish one element from another.

It will be understood that when an element is referred to as being “connected with” another element, the element can be directly connected with the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly connected with” another element, there are no intervening elements present.

A singular representation may include a plural representation unless it represents a definitely different meaning from the context.

Terms such as “include” or “has” are used herein and should be understood that they are intended to indicate an existence of several components, functions or steps, disclosed in the specification, and it is also understood that greater or fewer components, functions, or steps may likewise be utilized.

FIG. 1 is a block diagram of a mobile terminal in accordance with the present disclosure.

The mobile terminal 100 is shown having components such as a wireless communication unit 110, an input unit 120, a sensing unit 140, an output unit 150, an interface unit 160, a memory 170, a controller 180, and a power supply unit 190. Referring now to FIG. 1, the mobile terminal 100 is shown having wireless communication unit 110 configured with several commonly implemented components. It is understood that implementing all of the illustrated components is not a requirement, and that greater or fewer components may alternatively be implemented.

More specifically, the wireless communication unit 110 typically includes one or more modules which permit communications such as wireless communications between the mobile terminal 100 and a wireless communication system, communications between the mobile terminal 100 and another mobile terminal, communications between the mobile terminal 100 and an external server. Further, the wireless communication unit 110 typically includes one or more modules which connect the mobile terminal 100 to one or more networks.

To facilitate such communications, the wireless communication unit 110 includes one or more of a broadcast receiving module 111, a mobile communication module 112, a wireless Internet module 113, a short-range communication module 114, and a location information module 115.

Regarding the wireless communication unit 110, the broadcast receiving module 111 is typically configured to

5

receive a broadcast signal and/or broadcast associated information from an external broadcast managing entity via a broadcast channel. The broadcast channel may include a satellite channel, a terrestrial channel, or both. In some embodiments, two or more broadcast receiving modules **111** may be utilized to facilitate simultaneously receiving of two or more broadcast channels, or to support switching among broadcast channels.

The mobile communication module **112** can transmit and/or receive wireless signals to and from one or more network entities. Typical examples of a network entity include a base station, an external mobile terminal, a server, and the like. Such network entities form part of a mobile communication network, which is constructed according to technical standards or communication methods for mobile communications (for example, Global System for Mobile Communication (GSM), Code Division Multi Access (CDMA), CDMA2000 (Code Division Multi Access 2000), EV-DO (Enhanced Voice-Data Optimized or Enhanced Voice-Data Only), Wideband CDMA (WCDMA), High Speed Downlink Packet access (HSDPA), HSUPA (High Speed Uplink Packet Access), Long Term Evolution (LTE), LTE-A (Long Term Evolution-Advanced), and the like).

Examples of wireless signals transmitted and/or received via the mobile communication module **112** include audio call signals, video (telephony) call signals, or various formats of data to support communication of text and multimedia messages.

The wireless Internet module **113** is configured to facilitate wireless Internet access. This module may be internally or externally coupled to the mobile terminal **100**. The wireless Internet module **113** may transmit and/or receive wireless signals via communication networks according to wireless Internet technologies.

Examples of such wireless Internet access include Wireless LAN (WLAN), Wireless Fidelity (Wi-Fi), Wi-Fi Direct, Digital Living Network Alliance (DLNA), Wireless Broadband (WiBro), Worldwide Interoperability for Microwave Access (WiMAX), High Speed Downlink Packet Access (HSDPA), HSUPA (High Speed Uplink Packet Access), Long Term Evolution (LTE), LTE-A (Long Term Evolution-Advanced), and the like. The wireless Internet module **113** may transmit/receive data according to one or more of such wireless Internet technologies, and other Internet technologies as well.

In some embodiments, when the wireless Internet access is implemented according to, for example, WiBro, HSDPA, HSUPA, GSM, CDMA, WCDMA, LTE, LTE-A and the like, as part of a mobile communication network, the wireless Internet module **113** performs such wireless Internet access. As such, the Internet module **113** may cooperate with, or function as, the mobile communication module **112**.

The short-range communication module **114** is configured to facilitate short-range communications. Suitable technologies for implementing such short-range communications include BLUETOOTH™, Radio Frequency Identification (RFID), Infrared Data Association (IrDA), Ultra-WideBand (UWB), ZigBee, Near Field Communication (NFC), Wireless-Fidelity (Wi-Fi), Wi-Fi Direct, Wireless USB (Wireless Universal Serial Bus), and the like. The short-range communication module **114** in general supports wireless communications between the mobile terminal **100** and a wireless communication system, communications between the mobile terminal **100** and another mobile terminal **100**, or communications between the mobile terminal and a network where another mobile terminal **100** (or an external server) is

6

located, via wireless area networks. One example of the wireless area networks is a wireless personal area networks.

The location information module **115** is generally configured to detect, calculate, derive or otherwise identify a position of the mobile terminal. As an example, the location information module **115** includes a Global Position System (GPS) module, a Wi-Fi module, or both. If desired, the location information module **115** may alternatively or additionally function with any of the other modules of the wireless communication unit **110** to obtain data related to the position of the mobile terminal. As one example, when the mobile terminal uses a GPS module, a position of the mobile terminal may be acquired using a signal sent from a GPS satellite. As another example, when the mobile terminal uses the Wi-Fi module, a position of the mobile terminal can be acquired based on information related to a wireless access point (AP) which transmits or receives a wireless signal to or from the Wi-Fi module.

The input unit **120** includes a camera **121** for obtaining images or video, a microphone **122**, which is one type of audio input device for inputting an audio signal, and a user input unit **123** (for example, a touch key, a push key, a mechanical key, a soft key, and the like) for allowing a user to input information. Data (for example, audio, video, image, and the like) is obtained by the input unit **120** and may be analyzed and processed by controller **180** according to device parameters, user commands, and combinations thereof.

Such cameras **121** may process image frames of still pictures or video obtained by image sensors in a video or image capture mode. The processed image frames can be displayed on the display unit **151** or stored in memory **170**. In some cases, the cameras **121** may be arranged in a matrix configuration to permit a plurality of images having various angles or focal points to be input to the mobile terminal **100**. As another example, the cameras **121** may be located in a stereoscopic arrangement to acquire left and right images for implementing a stereoscopic image.

The microphone **122** is generally implemented to permit audio input to the mobile terminal **100**. The audio input can be processed in various manners according to a function being executed in the mobile terminal **100**. If desired, the microphone **122** may include assorted noise removing algorithms to remove unwanted noise generated in the course of receiving the external audio.

The user input unit **123** is a component that permits input by a user. Such user input may enable the controller **180** to control operation of the mobile terminal **100**. The user input unit **123** may include one or more of a mechanical input element (for example, a key, a button located on a front and/or rear surface or a side surface of the mobile terminal **100**, a dome switch, a jog wheel, a jog switch, and the like), or a touch-sensitive input, among others. As one example, the touch-sensitive input may be a virtual key or a soft key, which is displayed on a touch screen through software processing, or a touch key which is located on the mobile terminal at a location that is other than the touch screen. On the other hand, the virtual key or the visual key may be displayed on the touch screen in various shapes, for example, graphic, text, icon, video, or a combination thereof.

The sensing unit **140** is typically implemented using one or more sensors configured to sense internal information of the mobile terminal, the surrounding environment of the mobile terminal, user information, and the like. For example, the sensing unit **140** may alternatively or additionally include other types of sensors or devices, such as a

proximity sensor **141** and an illumination sensor **142**, a touch sensor, an acceleration sensor, a magnetic sensor, a G-sensor, a gyroscope sensor, a motion sensor, an RGB sensor, an infrared (IR) sensor, a finger scan sensor, a ultrasonic sensor, an optical sensor (for example, camera **121**), a microphone **122**, a battery gauge, an environment sensor (for example, a barometer, a hygrometer, a thermometer, a radiation detection sensor, a thermal sensor, and a gas sensor, among others), and a chemical sensor (for example, an electronic nose, a health care sensor, a biometric sensor, and the like), to name a few. The mobile terminal **100** may be configured to utilize information obtained from sensing unit **140**, and in particular, information obtained from one or more sensors of the sensing unit **140**, and combinations thereof.

The output unit **150** is typically configured to output various types of information, such as audio, video, tactile output, and the like. The output unit **150** is shown having a display unit **151**, an audio output module **152**, a haptic module **153**, and an optical output module **154**. The display unit **151** may have an inter-layered structure or an integrated structure with a touch sensor in order to facilitate a touch screen. The touch screen may provide an output interface between the mobile terminal **100** and a user, as well as function as the user input unit **123** which provides an input interface between the mobile terminal **100** and the user.

The audio output module **152** is generally configured to output audio data. Such audio data may be obtained from any of a number of different sources, such that the audio data may be received from the wireless communication unit **110** or may have been stored in the memory **170**. The audio data may be output during modes such as a signal reception mode, a call mode, a record mode, a voice recognition mode, a broadcast reception mode, and the like. The audio output module **152** can provide audible output related to a particular function (e.g., a call signal reception sound, a message reception sound, etc.) performed by the mobile terminal **100**. The audio output module **152** may also be implemented as a receiver, a speaker, a buzzer, or the like.

A haptic module **153** can be configured to generate various tactile effects that a user feels, perceive, or otherwise experience. A typical example of a tactile effect generated by the haptic module **153** is vibration. The strength, pattern and the like of the vibration generated by the haptic module **153** can be controlled by user selection or setting by the controller. For example, the haptic module **153** may output different vibrations in a combining manner or a sequential manner.

An optical output module **154** can output a signal for indicating an event generation using light of a light source. Examples of events generated in the mobile terminal **100** may include message reception, call signal reception, a missed call, an alarm, a schedule notice, an email reception, information reception through an application, and the like.

The interface unit **160** serves as an interface with various types of external devices that can be coupled to the mobile terminal **100**. The interface unit **160**, for example, may include any of wired or wireless ports, external power supply ports, wired or wireless data ports, memory card ports, ports for connecting a device having an identification module, audio input/output (I/O) ports, video I/O ports, earphone ports, and the like. In some cases, the mobile terminal **100** may perform assorted control functions associated with a connected external device, in response to the external device being connected to the interface unit **160**.

The memory **170** is typically implemented to store data to support various functions or features of the mobile terminal

100. For instance, the memory **170** may be configured to store application programs executed in the mobile terminal **100**, data or instructions for operations of the mobile terminal **100**, and the like. Some of these application programs may be downloaded from an external server via wireless communication. Other application programs may be installed within the mobile terminal **100** at time of manufacturing or shipping, which is typically the case for basic functions of the mobile terminal **100** (for example, receiving a call, placing a call, receiving a message, sending a message, and the like). It is common for application programs to be stored in the memory **170**, installed in the mobile terminal **100**, and executed by the controller **180** to perform an operation (or function) for the mobile terminal **100**.

The controller **180** typically functions to control overall operation of the mobile terminal **100**, in addition to the operations associated with the application programs. The controller **180** may provide or process information or functions appropriate for a user by processing signals, data, information and the like, which are input or output, or activating application programs stored in the memory **170**.

To drive the application programs stored in the memory **170**, the controller **180** may be implemented to control a predetermined number of the components mentioned above with reference to FIG. 1. Moreover, the controller **180** may be implemented to combinedly operate two or more of the components provided in the mobile terminal **100** to drive the application programs.

The power supply unit **190** can be configured to receive external power or provide internal power in order to supply appropriate power required for operating elements and components included in the mobile terminal **100**. The power supply unit **190** may include a battery, and the battery may be configured to be embedded in the terminal body, or configured to be detachable from the terminal body.

Some or more of the components may be operated cooperatively to embody an operation, control or a control method of the mobile terminal in accordance with embodiments of the present disclosure. Also, the operation, control or control method of the mobile terminal may be realized on the mobile terminal by driving of one or more application programs stored in the memory **170**.

FIG. 2 is a front view of a first state and a second state of the mobile terminal in accordance with an embodiment, and FIG. 3 is a rear view of the first state and the second state of the mobile terminal in accordance with an embodiment. FIGS. 2(a) and 3(a) are views showing the first state in which the mobile terminal is contracted, and FIGS. 2(b) and 3(b) are views showing the second state in which the mobile terminal is extended.

As shown in the figures, the mobile terminal **100** in the first state is in a contracted position, and has a smaller size than the mobile terminal **100** in the second state. In addition, the size of the display unit **151** positioned on the front of the mobile terminal **100** is also smaller than in the second state. The mobile terminal **100** in the first state may be extended in a first direction D1 to switch to the second state. In the second state, as shown in FIG. 2(b), the size of the mobile terminal **100** and the size of the display unit **151** positioned on the front of the mobile terminal **100** are larger than in the first state, while the size of the display unit **151** positioned on the rear of the mobile terminal **100** is reduced as shown in FIG. 3(b). That is, a part of the display units **151** positioned on the rear of the mobile terminal **100** in the first state moves to the front of the mobile terminal **100** in the second state.

In the following description, the direction in which the mobile terminal **100** and the display unit **151** thereof are extended or enlarged is defined as a first direction **D1**, and the direction in which the mobile terminal contracts or retracts or is reduced to switch from the second state to the first state is defined as a second direction **D2**. A direction perpendicular to the first and second directions **D1** and **D2** is defined as a third and fourth directions. Description will be made on the assumption that the first and second directions are horizontal directions and the third and fourth directions are vertical directions. However, depending on the arrangement of the mobile terminal **100**, the first and second directions may be vertical directions and the third direction may be a horizontal direction.

As such, a flexible display unit **151** which is bendable may be used as the display unit such that the position of the display unit may be varied. The flexible display unit **151** may be a display unit capable of maintaining a flat state like a conventional flat panel display and capable of warping, bending, folding, twisting, or rolling like paper. The flexible display unit **151** refers to a display which is manufactured on a thin and flexible substrate and is thus lightweight and robust as not to be easily broken. The flexible display unit according to the present disclosure may be bent in a specific direction, and may be arranged such that the curvature thereof may change in the first direction.

In addition, an electronic paper is a display technology to which properties of general ink are applied. The electronic paper may be different from the conventional flat panel display in using reflected light. The electronic paper may change information using a twisted ball or electrophoresis using a capsule.

In a state in which the flexible display unit **151** is not deformed (e.g., a state of having an infinite curvature radius, hereinafter referred to as a basic state), a display region of the flexible display unit **151** becomes flat. In a state in which the flexible display unit **151** is deformed by an external force from the basic state (e.g., a state of having a finite radius of curvature, hereinafter referred to as a deformed state), the display region may become a curved face. As shown, information displayed in the deformation state may be visual information output on the curved face. Such visual information is implemented by independently controlling light emission of sub-pixels arranged in a matrix. The sub-pixel refers to a minimum unit for implementing one color. When external force is applied to the flexible display unit **151**, the flexible display unit **151** may be deformed to switch from the default state, which is the flat state, to a bent state, which is not the flat state.

The flexible display unit **151** may be combined with a touch sensor to implement a flexible touch screen. When a touch is made on the flexible touch screen, the controller **180** (see FIG. 1) may perform control corresponding to such touch input. The flexible touch screen may be configured to detect the touch input in the deformed state as well as in the basic state.

The touch sensor detects the touch (or touch input) applied on the touch screen using at least one of various touch schemes such as a resistive film scheme, a capacitance scheme, an infrared scheme, an ultrasonic wave scheme, a magnetic field scheme, and the like.

As an example, the touch sensor may be configured to convert a change in pressure applied on a specific portion of the touch screen, capacitance generated at the specific portion, or the like into an electrical input signal. The touch sensor may be configured such that a touch object applying a touch on the touch screen may detect touched position and

area on the touch sensor, a pressure during the touch, a capacitance during the touch, and the like.

Further, the mobile terminal **100** may have a deformation detection means for detecting the deformation of the flexible display unit **151**. Such deformation detection means may be included in the sensing unit **140** (see FIG. 1).

The deformation detection means may be disposed in the flexible display unit **151** or a case (first to second frames **101** to **102** to be described later) to detect information related to the deformation of the flexible display unit **151**. In this connection, the information related to the deformation may include a direction in which the flexible display unit **151** is deformed, a degree of the deformation, a deformed position, a deformed time, an acceleration at which the deformed flexible display unit **151** is restored, and the like. In addition, the information related to the deformation may include various kinds of information that may be detected due to the bending of the flexible display unit **151**.

In addition, the controller **180** may change information displayed on the flexible display unit **151** or generate a control signal for controlling a function of the mobile terminal **100** based on the information related to the deformation of the flexible display unit **151** detected by the deformation detection means.

The deformation of the flexible display unit **151** may vary depending on the positions of the first frame **101** and the second frame **102**. As shown in FIG. 2, since the bending position on the flexible display unit **151** is determined according to the positions of the first frame and the second frame, the bending deformation position of the flexible display unit **151** and the area thereof positioned on the front may be calculated based on the positions of the first frame **101** and the first frame **102** in place of the deformation detection means of the flexible display unit **151**.

The state conversion (first or second state) of the flexible display unit **151**, i.e., the size change at the front and rear faces of the mobile terminal **100** of the display unit **151** based on the size change of the mobile terminal **100** may be performed manually by a force applied by the user, but may be not limited to such manual scheme. For example, when the mobile terminal **100** or the flexible display unit **151** is in the first state, the mobile terminal **100** or the flexible display unit **151** may be converted into the second state by the user or an application command without the external force applied by the user. As such, in order for the flexible display unit **151** to be automatically deformed without the external force, the mobile terminal **100** may include a driving unit **200**, which will be described later.

The flexible display unit **151** of the present disclosure is bent 180 degrees by rolling around a side portion of the mobile terminal **100** facing in the first direction. Accordingly, based on the side portion of the mobile terminal **100**, a part of the flexible display unit **151** is disposed on the front of the mobile terminal **100**, and the other part of the flexible display unit **151** is disposed on the rear of the mobile terminal **100**. For simplicity, the part of the flexible display unit **151** positioned on the front is called a front face, and the other part of the flexible display unit **151** positioned on the rear is called a rear face. As illustrated in FIG. 2, the mobile terminal may extend in the first direction or contract in the second direction opposite to the first direction. In this case, the area of the flexible display unit **151** positioned on the front changes. That is, the sizes of the front face and the rear face may be changed according to a change in the state of the mobile terminal.

The part of the flexible display unit **151** positioned on the front of the mobile terminal **100** may be immovably fixed to

11

the front surface of the first frame **101**, and the other part thereof positioned on the rear of the mobile terminal **100** may be movably arranged on the rear of the mobile terminal **100**.

In addition, the flexible display unit **151** may be rolled on or released at the side portion in the first direction of mobile terminal. Accordingly, the rear face of the display unit **151** moves, so that the size of the front face of the display unit **151** may be adjusted. Since the size of the flexible display unit **151** is determined and the flexible display unit **151** is formed of one continuous body, an area of rear face of the display unit **151** decreases as an area of the front face of the display unit **151** increases. Such a display unit **151** may be rolled in a second frame **102**, which is movable relative to a first frame **101** to be described later, more correctly, on one of sides of the second frame **102**. The display unit **151** may be withdrawn or pulled out from or inserted or pushed into the second frame **102** while being rolled in the second frame **102** along a moving direction of the second frame **102** to adjust the area of the display unit **151** on the front face of the mobile terminal **100**. Such operation will be described in more detail below along with other relevant components of the mobile terminal **100**.

Typically, an antenna is disposed in the case or the housing of the mobile terminal **100**, but a portion where the antenna is installed in the case or the housing may be limited because of the flexible display unit **151** that covers not only the front face of the mobile terminal **100** but also the rear face thereof. For this reason, the antenna may be implemented on the flexible display unit **151**. An antenna on display (AOD) is an antenna in which a transparent film is formed by stacking an electrode layer and a dielectric layer that have patterns engraved thereon, respectively. The antenna on display may be implemented thinner than an antenna implemented using a laser direct structuring (LDS) technology using a conventional copper nickel plating scheme, so that the antenna on display may not be exposed to the outside without affecting a thickness. In addition, the antenna on display may transmit and receive a signal directly to or from the display unit **151**. Accordingly, the antenna on display may be used in the mobile terminal **100** in which the display unit **151** is located on the both faces of the mobile terminal **100** as in the present disclosure.

FIGS. **4** and **5** are exploded perspective views of the mobile terminal in accordance with an embodiment. FIG. **4** is an exploded perspective view of the mobile terminal as viewed from the front side, and FIG. **5** is an exploded perspective view of the mobile terminal as viewed from the rear side.

The mobile terminal **100** of the present disclosure includes frames **101** and **102** in which components are mounted, and the frames **101** and **102** of the present disclosure may vary in size in the first direction as shown in FIG. **2**. One or more frames **101** and **102** move relative to each other, and sizes thereof may vary in the first direction. Electronic components are mounted in the frames **101** and **102**, and the flexible display unit **151** is located out of the frames **101** and **102**.

Since the mobile terminal **100** of the present disclosure includes the flexible display unit **151**, the flexible display unit **151** may be combined in a form surrounding front faces and rear faces of the frames **101** and **102**. The frame may include the first frame **101** and the second frame **102** moving in the first direction with respect to the first frame **101**. The first frame **101** and the second frame **102** include front portions, a rear portions, and side portions, respectively, and are coupled to each other.

12

First, the first frame **101** corresponds to a main body of the mobile terminal **100**, and may have a space between the first front portion **1011** and the first rear portion **1012** therein for accommodating various components. In addition, the first frame **101** may accommodate the second frame **102** movably coupled to the first frame **101** in such a space. More specifically, as shown in FIGS. **2** and **5**, the first frame **101** may include a first front portion **1011** disposed at a front portion of the mobile terminal **100** and supporting the front face of the display unit **151** and a first rear portion **1012** disposed at a rear portion of the mobile terminal and on which various components are mounted. A front deco **1016** may cover front edges of the first area **151a** to protect the edges of the first area **151a**.

The first front portion **1011** and the first rear portion **1012** may be spaced apart from each other at a predetermined spacing to define a predetermined space therebetween, and may be connected to each other by a first side portion **1013**. The first side portion **1013** may be integrally formed with the first rear portion **1012** or the first front portion **1011**. The camera **121**, the audio output module **152**, and the input/output terminal, the controller **180**, and the power supply unit **190** may be accommodated as components of the mobile terminal **100** in the space in the first frame **101**. For example, the controller **180** may be a circuit board **181** including a processor and an electronic circuit for controlling the operation of the mobile terminal, and the power supply unit **190** may be a battery **191** and related components. In addition, the driving unit **200** that controls the slide movement of the second frame **102**, which will be described later, may also be accommodated in the first frame **101**.

As described above, the display unit **151** has the continuous body, and thus, may be disposed on both the front face and the rear face of the mobile terminal **100** while being rolled in the mobile terminal **100**. The display unit **151** may include the front face positioned at the front face of the mobile terminal **100**, the rear face positioned at the rear face of the mobile terminal **100**, and the side face positioned between the front face and the rear face thereof and surrounding the side face of the mobile terminal. The front face and the rear face of the display unit **151** are flat, and the side face of the display unit **151** may form a curved face. The flexible display unit **151** may be damaged when being bent at an angle. Thus, the flexible display unit **151** may be formed to be bent with a predetermined curvature at the side face.

The display unit **151** may be divided into a fixed portion **151a** and **151b** and a variable portion **151c**. The fixed portion **151a** and **151b** means a portion fixed to the frame. Because of being fixed to the frame, the fixed portion **151a** and **151b** maintains a constant shape without changing a bending degree. On the other hand, the variable portion **151c** means a portion in which a bending angle or a position of the bent portion changes. The variable portion **151c** in which the position or bending angle of the bent portion changes requires a structure for supporting a rear face of the variable portion **151c** in response to the change.

The fixed portion **151a**, **151b** is coupled to the first frame of the display unit and is always positioned on the front face of the display unit to form a portion of the front face of the display unit. The variable portion **151c** includes a side face located at a side portion of the mobile terminal, and a position of the side face varies depending on the position of the second frame. Based on a side face, an area of a portion disposed on the front face of the display unit and an area of a portion disposed on the rear face of the display unit vary. That is, a portion of the variable portion **151c** may be the

13

front face and another portion of the variable portion **151c** may be the rear face based on the first and second states. The variable portion **151c** is positioned in the first direction with respect to the fixed portion **151a**, **151b** relative to the mobile terminal, and an end of the variable portion **151c** is bent toward the rear face of the mobile terminal and slides on the rear face of the second frame.

The end of the variable portion of the display unit is coupled with a slide frame that guides the variable portion to slide move on the rear face of the second frame, and the slide frame moves in the first direction at the same time as the second frame moves in the first direction. As a result, a moving distance of the slide frame with respect to the first frame is twice as a moving distance of the second frame with respect to the first frame. Further, as shown in FIG. 3, the first rear portion **1012** of the mobile terminal **100** includes an exposed rear portion **1015** that is exposed to the outside without being covered by the display unit **151** even in the first state. The physical input unit **120** for the manipulation of the mobile terminal **100** such as various buttons, switches, the camera **121**, and a flash, and the sensing unit **140** such as the proximity sensor **141** or a fingerprint sensor may be arranged on the exposed rear portion **1015**. The first rear portion **1012** except for the exposed rear portion **1015** may be covered by the display unit **151** in the first state as shown in FIG. 3(a), and may be exposed rearward in the second state as shown in FIG. 3(b).

In a conventional bar-shaped terminal, a display unit is provided only on a front face of the terminal. Therefore, a main camera is placed on a rear face of the terminal in order for the user to capture an object at an opposite side while looking through the display unit. On the other hand, an additional auxiliary camera is required to be disposed on the front face of the terminal in order for the user to capture himself or herself while viewing himself or herself through the display unit.

In the mobile terminal **100** of the present disclosure, on the other hand, the display unit **151** is arranged on both the front and rear of the mobile terminal **100**. Accordingly, when a user photographs himself, a portion of the display unit **151** positioned on the same surface as the camera **121**, that is, the rear face of the display unit **151** may be used. When the user takes a photograph of an object around the user, a portion of the display unit **151** on the side facing away from the camera **121**, that is, the front face of the display unit **151** may be used. For this reason, the mobile terminal **100** may take a photograph of the user or an object located around the user using one camera **121**. The camera may include a plurality of cameras having different angles of view, such as wide angle, ultra wide angle, and telephoto angle. Not only the camera but also a proximity sensor and an audio output unit may be disposed on the exposed rear portion **1015**, and an antenna may be installed on the rear portion **1015**. The rear portion **1015** may be used to protect the camera, the sensor, or the like on the exposed rear portion **1015** and not to deteriorate the exterior design. A portion of the rear portion **1015** corresponding to the camera **121** or the sensor **140** may be configured to be transparent, and the other portion thereof may have a predetermined pattern or color in consideration of design aspects without exposing internal parts.

The first side portion **1013** may extend along the edges of the first front portion **1011** and the first rear portion **1012** to surround the circumference of the first frame **101** and may define the appearance of the mobile terminal **100**. However, as mentioned above, the second frame **102** is accommodated in and movably coupled to the first frame **101**, and therefore a portion of the first frame **101** needs to be open to allow

14

movement of the second frame **102** relative to the first frame **101**. As an example, as best shown in FIG. 2, the second frame **102** may be movably coupled to a side of the first frame **101** facing in the first direction, and accordingly the first side portion **1013** may not be formed on the lateral surface facing in the first direction such that the lateral surface is open. Since the first side portion **1013** is exposed to the outside of the mobile terminal **100**, the interface unit **160** for connecting a power port or an ear jack or the user input unit **120**, such as a volume control button, may be disposed on the first side portion **1013**. When the first side portion **1013** contains a metal material, the first side portion **1013** may serve as an antenna.

The second frame **102** may include a second front portion **1021** disposed at the front portion of the mobile terminal **100** and a second rear portion **1022** disposed at the rear portion of the mobile terminal **100**. Like the first front portion **1011** and the first rear portion **1012** of the first frame **101**, the second front portion **1021** and the second rear portion **1022** may be formed of plate-shaped members that are generally flat. In addition, the second frame **102** also accommodates various components, and must not interfere with the components accommodated in the first frame **101** during the movement. Accordingly, the second front portion **1021** and the second rear portion **1022** may be coupled to each other in a state of being spaced apart from each other to define a predetermined space therebetween, and may have shapes that do not interfere with the components in the first frame **101**.

FIG. 6 is a side view of the mobile terminal as viewed from a third direction. FIG. 6 shows the first side portion **1013** of the first frame **101** and a second side portion **1023** of the second frame **102**. Since the flexible display unit **151** is positioned at an end of the second frame **102** facing in the first direction, the end of the second frame **102** facing in the first direction should not be exposed to the outside. An end of the second frame **102** facing in the second direction should be open so as not to interfere with the first frame **101**. In the first state, the second side portion **1023** of the second frame **102**, which is positioned on the side facing in the third direction (which refers to the upward or downward direction in the drawing or may include both the upward and downward directions), may not be exposed to the outside because it overlaps the first side portion **1013** of the first frame. However, in the second state, it may be exposed to the outside because the second frame **102** is drawn out.

In addition, the display unit **151** may be bent 180 degrees while being rolled in the second frame **102** to be disposed on both the front face and the rear face of the mobile terminal **100**. For such an arrangement of the display unit **151**, the second frame **102** may include a roller **210** rotatably disposed therein. The roller **210** may be disposed at any position inside the second frame **102**. However, the display unit **151** should be spread flat on the front face and the rear face of the mobile terminal **100** to provide a good quality screen to the user. Further, for such spread, a proper tension must be provided on the display unit **151**. In order to provide the proper tension, the roller **210** may be disposed at a first directional end of the second frame **102**. The roller **210** may extend in the third direction, and may be rotatably coupled to the second frame **102**.

The display unit **151** may be rolled around the roller **210** while being gently bent with a predetermined curvature. The flexible display unit **151** may include a first face on which a video is output and exposed to the outside and an inner face facing the frame at the opposite side. The roller **210** may be installed to rotate freely in the second frame **102** while being

15

in contact with the inner face of the display unit **151**. Accordingly, the roller **210** may actually move the display unit **151** in a lateral direction of the mobile terminal **100**, that is, in a direction perpendicular to a longitudinal direction. As will be described later, when the second frame **102** slides, because of the tension applied by the second frame **102**, the display unit **151** moves to the front face or the rear face of the mobile terminal **100** in different directions (i.e., the first direction **D1** or the second direction **D2**) relative to the second frame **102**. The roller **210** may guide such movement while rotating.

The roller **210** may be disposed adjacent to the end of the second frame **102** that faces in the first direction. A side frame **1024** may be disposed at the end of the second frame **102** facing in the first direction to prevent damage to the display unit **151** rolled around the roller **210**.

The side frame **1024** may extend in the longitudinal direction (the third direction) of the second frame **102** to cover the side portion facing in the first direction, thereby protecting the roller **210** and the display unit **151** rolled therearound.

An inner surface of the side frame may include a curved surface corresponding to the curvature of the side portion.

The side frame **1024** may substantially define the appearance of the mobile terminal **100** in cooperation with the first side portion **1013** of the first frame **101**. In addition, the side portion of the second frame **102** that faces in the second direction may be omitted to minimize interference with the components arranged in the first frame **101** during movement.

During the expansion and the contraction in such first and second directions **D1** and **D2**, the second frame **102** may overlap the first frame **101**, more precisely, the first front portion **1011** and the first rear portion **1012** thereof so as not to interfere with the first frame **101**. More specifically, the display unit **151** may be coupled to and supported by the first front portion **1011** of the first frame **101**, as described above. Accordingly, the display unit **151** does not need to be additionally supported by the second front portion **1021** of the second frame **102**. Rather, when the second front portion **1021** is interposed between the first front portion **1011** and the display unit **151**, the display unit **151** may be deformed or damaged because of friction with the second front portion **1021**, which is repeatedly moved. Thus, the second front portion **1021** may be disposed below the first front portion **1011**, or may be interposed between two first front portions **1011**. The second rear portion **1022** of the second frame **102** may be disposed rearward of the first rear portion **1012** of the first frame **101**. That is, the front face of the second rear portion **1022** may face the rear face of the first rear portion **1012**. In addition, the rear face of the first rear portion **1012** may be in contact with the front face of the second rear portion **1022** to stably support the movement of the second frame **102**. Because of such arrangement, the second rear portion **1022** may be exposed to the outside of the first frame, more precisely, of the first rear portion **1012**, and may be coupled to the display unit **151**.

In addition, the second frame **102** may extend and contract in the first and second directions **D1** and **D2** to change the size of the mobile terminal **100**, particularly, to extend or contract the front face of the mobile terminal **100**. Thus, the display unit **151** must move by such extended or reduced front face size to obtain the intended first and second states. However, when being fixed to the second frame **102**, the display unit **151** may not be moved smoothly to be adapted for the front face of the mobile terminal **100** that is extended

16

or contracted. For this reason, the display unit **151** may be movably coupled to the second frame **102**.

More specifically, the display unit **151** may include a first region **151a** disposed on the front of the mobile terminal **100**, a second region **151b** coupled to a slide frame **103** positioned on the rear of the mobile terminal **100**, and a third region **151c** located between the first region **151a** and the second region **151b** and bent around the roller **210**. The third region **151c** may move to the front or the rear according to change in the state of the mobile terminal **100**. The slide frame **103** may be formed of a plate-shaped member extending in the longitudinal direction (the third direction) of the mobile terminal **100**, and may be coupled to the second rear portion **1022** so as to be movable in the first and second directions **D1** and **D2**.

The first to third regions **151a**, **151b**, and **151c** may be connected to each other, and may form a continuous body of the display unit **151**. In addition, as described above, for the movement of the third region **151c** toward the front face or the rear face of the mobile terminal **100** depending on the moving direction of the second frame **102**, the first region **151a** may be fixed so as not to move to the front face of the mobile terminal **100**, and the second region **151b** may be provided to be movable on the rear face of the mobile terminal. Such configuration of the display unit **151** will be described in more detail below.

The first region **151a** may be disposed on the front face of the mobile terminal **100**, more specifically, the first frame **101**, that is, on the front face of the first front portion **1011**. The first region **151a** is fixed to the first frame **101**, that is, the front face of the first front portion **1011** so as not to be moved during the movement of the second frame **102**, and thus, the first region **151a** may always be exposed to the front face of the mobile terminal **100**.

The third region **151c** may be adjacent to the first region **151a** in a direction of a second end **151e**. The third region **151c** may extend into the second frame **102** and be rolled on the roller **210**. The third region **151c** may extend out of the second frame **102** and partially cover the second frame **102**, that is, the rear front of the second rear portion **1022**. Since the second frame **102**, i.e., the second rear portion **1022** is adjacent to the first frame **101**, i.e., the first rear portion **1012**, and the first and second frames **101** and **102** form the rear case of the mobile terminal **100**, it may be said that the third region **151c** is also disposed on the rear front of the first frame **101**.

The second region **151b** may be adjacent to the third region **151c** and may be disposed on the rear face of the mobile terminal **100**, more specifically, on the second frame **102**, that is, the rear face of the second rear portion **1022** thereof. The second region **151b** may be coupled to the slide frame **103** without being directly coupled to the second frame **102**.

As a result, the first region **151a** may be disposed on the front face of the mobile terminal **100** and may be always exposed to the front face regardless of the movement of the second frame **102**, and the second region **151b** may be disposed on the rear face of the mobile terminal **100** and may be always exposed to the rear face regardless of the movement of the second frame **102**. In addition, the third region **151c** may be disposed between the first and second regions **151a** and **151b**, and may be selectively placed on the front face or the rear face of the mobile terminal **100** depending on the moving directions **D1** and **D2** of the second frame **102**.

Because of such selective placement of the third region **151c**, the first rear portion **1012** of the first frame **101** may

17

be exposed to the outside of the mobile terminal **100** because the first rear portion **1012** is covered by the second and third regions **151b** and **151c** and the second rear portion **1022** of the display unit **151** in the first state, but, in the second state, the third region **151c** moves to the front face of the mobile terminal **100** and the second rear portion **1022** also moves in the first direction **D1**. In addition, the second front portion **1021** of the second frame **102** is hidden by the first front portion **1011** of the first frame **101** in the first state, but, in the second state, moves out of the first frame **101** to support the third region **151c** of the display unit **151** disposed on the front face of the mobile terminal **100**.

In order to prevent the second front portion **1021** from affecting the internal components during the slide movement, a separating plate **1017** may be further disposed rearward of the second front portion **1021** and fastened with the first front portion **1011**. The second front portion **1021** may move between the first front portion **1011** and the separating plate **1017** based on the slide movement of the second frame.

However, the third region **151c** may be rolled on the roller **210** and bent in the second frame **102**. When converting from the first state to the second state, the third region **151c** may extend from the second frame **102** to the front face of the mobile terminal **100** while being rolled on the roller **210** in one direction. On the other hand, when converting from the second state to the first state, the third region **151c** may be retracted from the front face of the mobile terminal **100** to the second frame **102** while being rolled on the roller **210** in the opposite direction, and at the same time, may return to the rear face of the mobile terminal **100** from the second frame **102**.

A specific location of the foldable mobile terminal in a form of being spread like a book is easily damaged because only the specific location is folded repeatedly. On the other hand, the deformed portion of the flexible display unit **151**, that is, a portion rolled on the roller **210**, may vary based on the first and second states of the mobile terminal **100**, that is, the movement of the second frame **102**. Accordingly, the mobile terminal **100** of the present disclosure may significantly reduce deformation and fatigue repeatedly applied to a specific portion of the display unit **151**, thereby preventing damage to the display unit **151**.

Based on the above-described configuration, overall operations of the mobile terminal **100** will be described as follows. As an example, the state conversion may be performed manually by the user, and an operation of the mobile terminal **100** during such manual state conversion will be described. However, operations of the first to third frames **101** to **103** and the display unit **151**, which will be described below, may be performed in the same manner when a power source other than a user's force is used, for example, when the driving unit **200** to be described below is applied.

A rear face cover **1025** may be further disposed on a rear face of the second rear portion **1022** such that the rear face of the display unit positioned on the rear face of the mobile terminal **100** is not exposed to the outside. The rear face of the display unit may be used in the first state when the rear face cover **1025** uses a transparent material, and the rear face of the display unit may be covered such that the movement of the slide frame **103** is not exposed when the rear face cover **1025** uses an opaque material. That is, the second region and the third region of the slide frame **103** and the display unit **151** may move in the first direction and in the second direction in a space between the second rear portion **1022** and the rear face cover **1025**.

18

FIG. 7 is a view showing a driving unit **200** of the mobile terminal **100** in accordance with an embodiment. FIG. 7(a) shows the first state and FIG. 7(b) shows the second state. The mobile terminal **100** of the present disclosure may be switched between the states in a manner in which a user manually pulls the second frame **102** in the first direction **D1** or pushes the same in the second direction **D2** with respect to the first frame **101**. However, in the manual method, applying excessive force to the body of the mobile terminal **100** may damage the mobile terminal **100**. Accordingly, a driving unit **200** employing a motor **201** may be further provided to cause the second frame **102** to stably move without distortion.

As the motor **201**, a motor **201** configured to provide rotational force as shown in FIG. 7, or a linear motor **201** configured to make linear motion may be used. The motor **201** configured to provide the rotational force should have a large diameter to provide large force. Two motors **201** may be used as shown in FIG. 7 to provide driving force of a predetermined magnitude or more in the limited space of the mobile terminal **100** without increasing the thickness.

If the second frame **102** is moved excessively fast, damage or malfunction may occur. Accordingly, a planetary gear configured to decrease the speed of the motor **201** to ensure movement at a stable speed may be further provided. The planetary gear **202** serves to amplify or attenuate the number of revolutions of the motor **201** using a plurality of disc gears having different numbers of teeth. The motor **201** may be fixed to the first frame **101** as shown in FIG. 7(a). The position of the motor **201** is fixed even when the second frame **102** moves in the first direction to switch the mobile terminal **100** to the second state, as shown in FIG. 7(b).

Since the second frame **102** linearly moves with respect to the first frame **101** in the first direction or the second direction, rack and pinion gears configured to convert the rotational force of the motor **201** into linear motion may be used. A pinion gear to receive the rotational force of the motor **201** may be arranged to engage with a rack gear **205** composed of teeth continuously arranged in the first direction. The pinion gear may be fixed to the first frame **101** together with the motor **201** and the rack gear **205** may be positioned on the second frame **102**. Alternatively, the rack gear **205** may be positioned on the first frame **101**, and the motor **201** and the pinion gear may be arranged on the second frame **102**. Since the motor **201** holds the pinion gear such that the pinion gear does not rotate, the second frame **102** may maintain the first state and the second state. However, when large external force is applied, the second frame **102** may be displaced as the pinion gear rotates.

A stopper (not shown) configured to fix the positions of the second frame **102** or the rack gear **205** and the first frame **101** may be further provided to fix the mobile terminal **100** in the first state or the second state. When electric current flows through the motor **201** to drive the motor **201**, the stopper may be released to allow the movement of the second frame **102**. When power is not applied to the motor **201** and thus the motor **201** does not rotate, the first frame **101** and the second frame **102** may be fastened such that the positions thereof are fixed.

When a pair of driving units **200** is symmetrically disposed in the vertical direction (the third direction), stable movement may be made. However, to arrange a battery or the like, the driving unit **200** should be arranged biased to one side in consideration of the limited mounting space of the mobile terminal **100** as shown in FIG. 7(a). According to such asymmetric arrangement of the driving unit **200**, the second frame **102** may be distorted during movement due to

19

a difference in movement speed between the upper end portion and the lower end portion. To address this issue, a linear guide **230** may be further provided.

The linear guide **230** may be disposed at both ends of the mobile terminal **100** facing in the third direction, that is, on the upper and lower sides of the mobile terminal **100**, in order to supplement the function of one driving unit **200** biased to one side in the third direction. The linear guide **230** may include a guide rail **231** extending in the first direction and a guide block **232** configured to move along the guide rail **231**. The guide rail **231** may be disposed on the first frame **101** and the guide block **232** may be disposed on the second frame **102**, or vice versa. In this embodiment, the guide rail **231** may be disposed on the second frame **102** to cover the upper and lower sides of the extended portion of the second frame **102** in the second state.

After the guide block **232** is coupled to the first frame **101** and the guide rail **231** is coupled to the second frame **102**, the guide block **232** and the guide rail **231** may be slidably fastened to each other. However, for convenience of the fastening, the guide block **232** and the guide rail **231** fastened to each other. Then, the guide block **232** may be first fixed to the first frame **101**, and then the second frame **102** may be coupled to the guide rail **231**.

The guide block **232** may be provided with a guide groove into which the guide rail **231** is inserted. Alternatively, the guide rail **231** may be provided with a rail groove into which a portion of the guide block **232** is inserted. The fastening portions of the guide rail **231** and the guide block **232** may be formed to be bumpy. Accordingly, movement in the first direction or the second direction may be made without displacement in the thickness direction of the mobile terminal **100**. In order to reduce friction between the guide block **232** and the guide rail **231**, a self-lubricating member having high wear resistance and low friction resistance, such as a bearing or polyoxymethylene (POM), may be added to the inside of the guide groove.

FIG. **8** is a cross-sectional view taken along lines A-A and B-B in FIG. **2**. As illustrated in FIG. **2**, when the second frame **102** switches to the second state by moving in the first direction, the third region **151c** positioned on the rear side moves to the front, and thus a structure to support the rear surface of the third region **151c** moved to the front is required. The second front portion **1021** positioned on the front surface of the second frame **102** may be positioned on the rear surface of the third region **151c** in the second state. However, in the first state, the second front portion **1021** is disposed to overlap the first front portion **1011** of the first frame **101**, and accordingly the first front portion **1011** and the second front portion **1021** form a step. A boundary is formed between the first region **151a** and the third region **151c** of the flexible display unit **151** by the step formed by the first front portion **1011** and the second front portion **1021**. A rolling hinge **104** may be used as a support structure to fill the gap between the second front portion **1021** and the third region **151c** of the flexible display unit **151**.

The rolling hinge **104** may be positioned on the rear surface of the flexible display unit **151**, and have a thickness corresponding to the gap between the second front portion **1021** and the flexible display unit **151** in the second state. As shown in FIG. **8(a)**, in the first state, the rolling hinge **104** is rolled around the roller **210** and is positioned on the lateral side and rear side of the mobile terminal **100**. The flexible display unit **151** and the rolling hinge **104** may be positioned between the second rear portion of the second frame **102** and a rear cover **1025** provided to cover the rear face of the display unit **151**. As shown in FIG. **8(b)**, when switch to the

20

second state occurs, the rolling hinge **104** may move to the front and the rolling hinge **104** may be positioned on the front portion of the second frame **102**.

The third region **151c** of the display unit **151** in which the rolling hinge **104** is positioned is a portion where bending deformation occurs when switch from the first state to the second state occurs. Accordingly, the rolling hinge **104** may be deformed according to deformation of the third region **151c**. Here, the rolling hinge **104** is required to have a predetermined stiffness to maintain the flat state when the flexible display unit **151** is positioned on the front or rear of the mobile terminal. That is, the rolling hinge **104** needs a structure capable of maintaining the flat state in the third direction and performing bending deformation in the first direction.

FIG. **9** is a view illustrating a display unit **151** and a rolling hinge **104** of the mobile terminal **100** in accordance with an embodiment. The rolling hinge **104** may include multiple support bars **1041** extending in the third direction. The multiple support bars **1041** may be arranged side by side in the first direction and spaced apart from each other by a predetermined distance. Accordingly, even when the flexible display unit **151** is rolled around the roller **210** and is thus bent, interference between the support bars **1041** may be avoided. The support bars **1041** may be implemented with an injection molding material having a predetermined thickness for stiffness, and may include materials such as SUS or ferrosilicon (FeSi).

The multiple support bars **1041** may be directly attached to the rear surface of the display unit **151**. However, this operation may take a long time and produce a lot of defects, resulting in poor productivity. In addition, directly processing the display unit **151** is highly likely to damage the display unit **151**. Therefore, a rolling sheet **1045** to fix the multiple support bars **1041** may be further provided. The rolling sheet **1045** may include a metal material, and may employ a superelastic material that is bending-deformable and capable of recovering the flat state after the bending deformation. For example, a superelastic metal sheet such as a thin STS sheet of 0.05 mm or less may be used. An adhesive tape may be attached to both surfaces of the rolling sheet **1045** to bond the rolling sheet **1045** to the support bars **1041** and bond the rear surface of the display unit **151** to the rolling sheet **1045**.

The rolling sheet **1045** may be provided with a kerf pattern in which multiple grooves extending in the third direction are formed in the first direction. The grooves in the kerf pattern may be formed between the multiple support bars **1041**. The grooves may be formed on a surface of the rolling sheet **1045** to which the support bars **1041** are bonded. The kerf pattern may be formed in a wedge shape that is formed by being gradually narrowed from the surface portion of the rolling sheet **1045**.

Instead of the rolling sheet **1045**, an elastic material such as silicone may be disposed between the support bars **1041** to join neighboring support bars **1041**. In this case, the angle between the support bars **1041** may be varied. The elastic connector may be bent at a position corresponding to the roller **210**. When positioned on the front or rear of the mobile terminal, the elastic connector may be unfolded such that the support bars **1041** are disposed forming a flat surface.

The support bars **1041** may form a flat surface corresponding to the rear surface of the display unit **151**. Alternatively, as shown in FIG. **8(b)**, the support bars **1041** may be formed in a shape having a predetermined curvature. The curved support bars **1041** may closely contact the curved

21

surface of the roller **210** when the rolling hinge **104** is rolled around the roller **210**. Alternatively, one surface of the support bars **1041** in contact with the display unit **151** maintains a flat state, and the other surface thereof on the opposite side may include a curved surface corresponding to the curvature of the roller **210**. In this case, the support bars **1041** may be thick at the ends thereof facing in the first and second directions and have the thinnest portion in the middle thereof.

The rolling hinge **104** may be disposed at a position corresponding to the third region **151c** and is rolled and bent around the roller **210**. Thus, the rolling hinge **104** may span over the front and rear surfaces. The rolling hinge **104** is connected to the first front portion **1011** of the first frame **101** on the front side and connected to the slide frame **103** on the rear side. In order for the flexible display unit **151** to form a continuous surface without a step, the first front portion **1011** of the first frame **101** positioned on the rear surface of the first region **151a**, the slide frame **103** positioned on the rear surface of the second region **151b**, and the rolling hinge **104** positioned on the rear surface of the third region **151c** may be arranged such that the surfaces thereof in contact with the display unit **151** are at the same height. In particular, since the slide frame **103** moves on the rear of the mobile terminal **100** and moves in the same space as the rolling hinge **104**, the rolling hinge **104** may have a thickness corresponding to the thickness of the slide frame **103**.

FIG. **10** is a view illustrating an issue raised in an extendable mobile terminal **100**. The figure shows the end of the second frame **102** facing in the third direction when the second frame **102** positioned overlapping the first frame **101** is extended as switching from the first state to the second state occurs. Since the display unit **151** contains a flexible material, the display unit **151** may be bent at a position corresponding to the roller **210**. However, when the display unit **151** is positioned on the front of the mobile terminal **100** in the second state, it should be maintained in a flat state. Since the rolling hinge **104** is not fixed to the second frame **102**, the display unit **151** may sag downward as shown in FIG. **10(a)** or rise upward as shown in FIG. **10(b)**.

The sagging issue of the display unit **151** (in FIG. **10(a)**) may be addressed when the support bars **1041** of the rolling hinge **104** described above support the rear surface of the third region **151c** of the display unit **151** and the front portion of the second frame **102** supports the support bars **1041**. However, in order to address an issue that the display unit **151** moved to the front is not arranged flat but is separated from the second frame **102** as shown in FIG. **10(b)**, the rolling hinge **104** and the second frame **102** may further include a slide structure **1043**, **1027**. The slide structures **1043** and **1027** may fasten the rolling hinge **104** (the display unit **151**) so as not to be separated from the second frame **102** in the thickness direction of the mobile terminal **100** while allowing the movement thereof only in the first direction or the second direction.

FIGS. **11** to **13** are views showing the slide structures **1043** and **1027** provided on the rolling hinge **104** and the second frame **102**. FIG. **11** is a cross-sectional view taken along lines C-C and D-D in FIG. **2**, FIG. **12** is a cross-sectional view taken along lines E-E and F-F in FIG. **10**, and FIG. **13** is a view showing an expanded side portion **1042** and a guide hook **1043** of the mobile terminal **100** in accordance with an embodiment.

As shown in FIG. **13**, the rolling hinge **104** may include a slide hook **1043** protruding toward the second side portion **1023** positioned on the side of the second frame **102** facing in the third direction. As shown in FIG. **11**, the second side

22

portion **1023** of the second frame **102** may include a slide rail **1027** into which the slide hook **1043** is movably inserted. FIG. **11** shows the slide rail **1027** formed on the second side portion **1023** of the second frame **102**. The slide rail **1027** may include a pair of horizontal portions **1027a** and **1027b** arranged side by side in a thickness direction and a curved portion **1027c** connecting ends of the horizontal portions **1027a** and **1027b** facing in the first direction. The pair of horizontal portions **1027a** and **1027b** and the curved portion **1027c** may form a U shape, and the slide hook **1043** may move along the slide rail **1027**.

In the first state, the slide hook **1043** may be positioned in the horizontal portion **1027a**, **1027b** positioned on the rear side, as shown in FIG. **11(a)**. In the second state, the slide hook **1043** may be moved to the horizontal portion **1027a**, **1027b** positioned on the front side, as shown in FIG. **11(b)**. When moved from the rear horizontal portion **1027a**, **1027b** to the front horizontal portion **1027a**, **1027b**, the slide hook **1043** may pass through the curved portion **1027c**. When the slide hooks **1043** pass through the curved portion **1027c**, the rolling hinge **104** may be bent and the space between the slide hooks **1043** may be narrowed.

The slide rail **1041** along which the slide hook **1043** moves may guide the movement of the slide hook **1043** and the movement of the slide frame **103** simultaneously. Since the slide frame **103** is also disposed adjacent to the support bars, a slide hook **1033** protruding from the slide frame **103** may be further provided.

FIG. **12(a)** shows the second frame **102** in the first state in which the first frame **101** and the second frame **102** overlap each other, and FIG. **12(b)** shows the second frame **102** moved from the first frame **101** in the first direction in the second state. As shown in FIG. **12(a)**, the support bars **1041** and the slide hooks **1043** are positioned on the rear in the first state. In the second state, as shown in FIG. **12(b)**, the support bars **1041** and the slide hooks **1043** are positioned on the front. The distance from the front face of the display unit **151** to the front horizontal portions **1027a** and **1027b** may be equal to the distance from the rear face of the display unit **151** to the rear horizontal portions **1027a** and **1027b**.

When large friction occurs during movement of the slide hooks **1043** along the slide rail **1027**, a large load may be applied to the driving unit **200**, thereby generating frictional noise or affecting the durability of the product. A self-lubricating member having high abrasion resistance and low friction resistance, such as polyoxymethylene (POM), may be used to facilitate the movement of the slide hooks **1043** on the slide rail **1027**. By forming the slide hooks **1043** in a cylindrical shape, the area thereof in contact with the slide rail **1027** may be minimized.

As illustrated in FIG. **13**, a slide roller **1044** fitted onto the slide hook **1043** may be further provided. The slide roller **1044** may be fitted onto the cylindrical slide hook **1043** to rotate about an axis extending in a direction in which the slide hook **1043** protrudes. The slide roller **1044** may rotate when the slide hook **1043** moves along the slide rail **1027**, thereby reducing friction generated between the slide roller **1044** and the slide rail **1027**.

A bearing structure may help to reduce the friction, but it may increase the overall size. For this reason, a simple ring-shaped slide roller **1044** may be used. A fastening groove **1043a** and a fastening protrusion **1044a** may be further provided to prevent the slide roller **1044** from being separated from the slide hook **1043**. When the fastening protrusion **1044a** is inserted into the fastening groove **1043a**, the fastening groove **1043a** formed on the outer circumferential surface of the slide hook **1043** in a ring

23

shape and the ring-shaped fastening protrusion **1044a** protruding from the inner surface of the slide roller **1044** are fastened to each other.

The second side portion **1023** of the second frame **102** is disposed overlapping the first side portion **1013** of the first frame **101** in the first state, and is exposed to the outside in the second state. Since the second side portion **1023** is positioned inside the first side portion **1013** of the first frame **101** in the first state, it may be difficult to connect the interface unit **160**, the user input unit **123**, the audio output unit **152**, the antenna, and the like, which are positioned on the first side portion **1013**, to the printed circuit board **181**, which is positioned inside.

The second side portion **1023** may include an opening **1023a** formed by omitting at least a part of the second side portion **1023** to connect the components positioned on the first side portion **1013** to the printed circuit board **181** positioned inside the first frame **101**. Through the opening **1023a**, a connector may be arranged between the components positioned on the first side portion **1013** and the printed circuit board. The opening **1023a** may have a shape elongated in the first direction to prevent interference between the connector and the second side portion **1023** of the second frame **102** even when the second frame **102** is moved.

However, in order to prevent the inside of the mobile terminal **100** from being exposed through the opening **1023a** in the second state, an expanded side portion **1042** may be formed at an end of the support bar **1041** to cover the opening **1023a** in the second state. The expanded side portion **1042** may have a larger area than the cross section of the support bar **1041** at the end of the support bar **1041**. The slide hook **1043** may be formed on a first expanded side portion **1042a** extending in a direction away from the display unit **151**. When the slide hook **1043** is disposed parallel to the support bar **1041**, the second side portion **1023** of the second frame **102** should be extended adjacent to the display unit **151**, which may make it difficult to form the opening **1023a**. Accordingly, the slide hooks **1043** may be formed at positions spaced apart from the display unit **151** by a predetermined distance. The first expanded side portion **1042a** may be formed in a fan shape to have a width that is reduced as the first expanded side portion **1042a** extends in a direction away from the display unit **151**. FIG. 13 is a view showing an embodiment of the rolling hinge **104** of the mobile terminal **100**. Since the rolling plates at positions on the curved portion **1027c** of the slide rail **1027** spaced apart from the display portion **151** are disposed adjacent to each other, the first expanded side portion **1042a** may be formed in a fan shape to prevent the rolling plates from overlapping each other.

When the rolling hinge **104** is positioned in the curved portion **1027c** of the slide rail **1027**, since neighboring expanded side portions **1042** are close to each other in a portion spaced apart from the display unit **151**, i.e., at a position in which the slide hook **1043** is disposed. The expanded side portion **1042** may have the shape of a fan where its width decreases as the expanded side portion **1042** is farther away from the support bar **1041** to prevent an overlap between first expanded portions **1042a**. The expanded side portion **1042** may also extend in the left-right direction, and may thus cover a space between neighboring support bars **1041**. The expanded side portion **1042** may further include a second expanded side portion **1042b** forming a step with respect to the first expanded side portion **1042a** as shown in FIG. 13. The second extension side portion **1042b** may be disposed to overlap the first expanded

24

side portion **1042a** adjacent thereto to cover the space between the support bars **1041**. The second expanded side portion **1042b** may have a size corresponding to the thickness-wise size of the opening **1023a** formed in the second side portion **1023** of the second frame **102**. The second expanded side portion **1042b** may be formed to be shorter than the first expanded side portion **1042a** on the side spaced apart from the display unit **151**. The second expanded side portion **1042b** may be formed to be shorter than the first expanded side portion **1042a** to prevent the second expanded side portions **1042b** from overlapping each other when the rolling plates pass through the curved portion **1027c** of the slide rail **1027**. The second expanded side portion **1042b** may be formed to have a size corresponding to the size of the opening **1023a** to cover the space between the first expanded side portions **1042a** to prevent the inside of the second frame **102** from being exposed to the outside. FIG. 14 is a view showing the slide frame **103** and the second frame **102** of the mobile terminal **100**. Like the rolling hinge **104**, the slide frame **103** is positioned on the rear surface of the display unit **151** and moves together with the display unit **151**. The rolling hinge **104** moves to the back and front of the mobile terminal **100**, but the slide frame **103** moves only on the back of the mobile terminal **100**, that is, on the second rear portion **1022** of the second frame **102**. A structure to guide the slide frame **103** to linearly move in the first direction or the second direction without distortion is needed. A guide slot extending in the first direction may be formed in the rear surface of the second frame **102** and the slide frame **103** may be configured to move along the guide slot.

However, since the second side portion **1023** of the second frame **102** includes the slide rail **1027** extending in the first direction as described above, the movement of the slide frame **103** may be guided through the slide rail **1027**, into which the slide hook **1043** of the rolling hinge **104** is inserted, without forming a separate guide rail. As shown in FIG. 15, slide hooks **1033** may be formed at both ends of the slide frame **103** to guide the slide frame **103** to move along the slide rail **1027**. The slide hooks **1033** of the slide frame **103** may have a shape similar to that of the slide hook **1043** of the rolling hinge **104**. The slide hooks **1033** protrude from the display unit **151** in a direction away from the display unit **151**. However, since the slide hooks **1033** of the slide frame **103** do not move to the front, they move only on a first horizontal portion.

Since the slide frame **103** and the rolling hinge **104** move along the slide rail **1027**, movement in the first direction and movement in the second direction may be guided stably, and the issue of separation of the display unit **151** from the second frame **102** may be addressed. However, since the slide rail **1027** is positioned at the end facing in the third direction, the middle portion of the slide frame **103** in the third direction may be separated from the second frame **102**. To prevent the separation at the middle portion of the slide frame **103** while both ends of the slide frame **103** are fixed to the frame **102**, a magnetic material may be used.

FIG. 15 is a graph showing phase distortion in the mobile terminal **100**, and FIG. 16 is a conceptual view for explaining causes of the phase distortion of FIG. 15.

The flexible display unit **151** may include a portion (fixed portion) attached to the first frame **101** and a portion (variable portion) not attached to the first frame **101** or a bending portion. The first region **151a** is fixed to the first frame **101**, but the second region **151b** is not fixed to the first frame **101** so that it is capable of bending or moving to the front and rear side.

25

In FIG. 15, ①, ②, and ③ respectively show phase distortion in the display unit 151 at the top, middle, and bottom of the fixed portion, and ④, ⑤, and ⑥ respectively show phase distortion in the display unit 151 at the top, center, and bottom of the variable portion. Since the fixed portion is fixed to the first frame 101, the amount of phase distortion is relatively low. In particular, since the center part is more stably attached to the first frame 101 than the top and bottom parts, the amount of phase distortion at the center part is lower than the top and bottom parts.

However, since an end in the second direction, i.e., the left end in the drawing is attached to the first frame 101 while being bent as shown in FIG. 16, it may have a repulsive force to the flatness of the display unit 151. As a result, phase distortion may occur at the end in the second direction. In addition, the end of the fixed portion facing in the first direction is adjacent to the variable portion, phase distortion may occur thereat due to the deformation of the variable portion.

The rear surface of the variable portion is supported by the rolling hinge 104. However, since the rolling hinge 104 has a deformable structure unlike the frames, the phase distortion at the fixed portion may increase. The phase distortion may increase at a point between the fixed and variable portions where the support structure (e.g., frame or rolling hinge) is partially omitted on the rear surface thereof. Further, the phase distortion may also increase at a point where the variable portion is rolled on the roller 210.

Referring to FIG. 16, the phase distortion at the end of the fixed portion facing in the second direction, the point between the fixed and variable portions, and the point where the variable portion is rolled on the roller 210 may be convex outward, whereas the phase distortion at the center of the fixed portion attached to the first frame 101 may be reverse to the former distortion.

The phase distortion at the center of the variable portion is similar to that of the fixed portion. A line distinguishing between the support bar 1041 and a portion combined with the support bar 1041 of the rolling hinge 104 on the rear surface may appear on the display unit 151.

To minimize the phase distortion, the display unit 151 needs to be in a tight state. To reduce vertical distortion in the display unit 151, the variable portion of the display unit 151 should be flat.

Since the end of the third region (variable portion) facing in the vertical direction (third direction) is fixed by the liner guide 230, there may be no waviness. However, since there may be waviness in the center thereof facing in the third direction, a rivet rail 1029 or slide rivets 1049 and 1039 may be used to prevent from the second area 151b or the third area 151c from separated from the second frame 102.

FIG. 17 is a view showing coupling of the roller 210 and the rolling hinge of the mobile terminal. The rolling hinge 104 includes the plurality of support bars 1041, and the support bars 1041 extend in the third direction and are arranged side by side in the first direction. In order to avoid interference between neighboring support bars 1041 when the display unit 151 is wound around the roller 210, the support bars 1041 are arranged to be spaced apart from each other by a predetermined distance as shown in FIG. 17, or may have a cross-section in which width decreases in a direction away from the display unit 151.

When the plurality of support bars 1041 are spaced apart from each other or the support bar 1041 decreases in the width toward an end thereof, the plurality of support bars 1041 may serve as rack gears. Teeth 2141 having protrusions,

26

each of which is inserted between two neighboring support bars 1041, may be formed on the roller 210. The roller 210, which has a smooth surface, may not be in close contact with the rolling hinge 104 and may be pushed, causing the display unit 151 to lift.

In the present disclosure, because the teeth 2141 of the roller 210 and the support bars 1041 of the rolling hinge 104 are engaged with each other and moved, the rotation of the roller 210 and the movement of the display unit 151 may be synchronized without misalignment.

A driving force may be provided to the roller 210 itself, but because the motor 201 of the aforementioned driving unit 200 is located on the first frame 101, there is a problem in that a separate motor is required in order to provide the driving force to the roller 210 located on the second frame 102. Because a mounting space of the mobile terminal 100 is limited to have the additional motor, in the present disclosure, an elastic member 213 may be added to the roller 210 to provide a force for assisting the movement of the display unit 151.

FIG. 18 is an exploded perspective view of the roller 210 of the mobile terminal 100, and FIG. 19 is a view showing an order of fastening each component of the roller 210. The roller 210 of the present disclosure includes a fixed shaft 211, a driving shaft 212, a coil spring 213, and a rotating cylinder 214, and is located in a seating portion 1028 (see FIG. 21) of the roller 210 located at an end in the first direction of the second frame 102. The fixed shaft 211 is fixed to the seating portion 1028 and does not rotate, and the drive shaft 212 is rotatably coupled to the fixed shaft 211.

One end 2111 of the fixed shaft is fixed to the seating portion 1028, and the seating portion 1028 of the roller 210 may include a fixing hole 10281 into which a screw passing through one end 2111 of the fixed shaft is inserted. The other end 2112 of the fixed shaft extends long in a direction of the driving shaft 212. One end 2121 of the driving shaft is rotatably coupled to the second frame 102.

The seating portion 1028 of the roller 210 may include a bearing 10282 such that one side 2121 of the driving shaft is rotatably seated. The other end 2122 of the driving shaft is disposed to face the other end 2112 of the fixed shaft and is rotatably coupled to the fixed shaft 211. The other end 2122 of the driving shaft may have a cylindrical shape surrounding the other end 2112 of the fixed shaft.

The rotating cylinder 214, which forms an outer surface of the roller 210 and has the teeth 2141 formed on an outer surface thereof and engaged with the support bars 1041, is coupled to the driving shaft 212 and rotates together with the driving shaft 212. The rotating cylinder 214 may be located on a circumference of the coil spring 213 and rotatably fastened with the driving shaft 212. One side of the rotating cylinder 214 may be coupled to the driving shaft 212, and the other side thereof may be formed in a shape surrounding a circumference of the other end 2122 of the driving shaft.

The coil spring 213 having one end coupled to the fixed shaft 211 and the other end coupled to the driving shaft 212 provides a rotational force to the roller 210. There is a form of the coil spring 213 providing elasticity in an axial direction, but a torsion spring that provides the elasticity to restore the original number of turns when the number of turns changes may be used as the coil spring 213 of the present disclosure.

Ends of the coil spring 213 may be bent so as to extend in a radial direction to be engaged with the fixed shaft 211 and a rotation axis, and the fixed shaft 211 and the driving shaft 212 may include coupling grooves into which the ends of the coil spring 213 are fastened. The coupling grooves

27

may extend in a direction transverse to the rotation axis corresponding to the extension direction of the ends of the coil spring 213.

For the order of coupling, one side of the coil spring 213 is fixed to the fixed shaft 211 as shown in (a) in FIG. 19. In order to insert the coil spring 213 into the fixed shaft 211 in a direction of the rotation axis, a first coupling groove 2113 of the fixed shaft 211 may extend to the other end of the fixed shaft 211. After inserting one side of the coil spring 213 into the first coupling groove 2113 of the fixed shaft 211, the rotating cylinder 214 is coupled to be located at an outer circumference of the coil spring 213 as shown in (b) in FIG. 19.

Next, as shown in (c) in FIG. 19, the driving shaft 212 is inserted in the direction of the rotation axis such that the other end 2132 of the coil spring and a second coupling groove 2123 of the driving shaft 212 are coupled to each other. Like the first coupling groove 2113, the second coupling groove 2123 may also extend to the other end 2122 of the driving shaft such that the coil spring 213 and the driving shaft 212 may be coupled to each other in the direction of the rotation axis. The other end 2132 of the coil spring is inserted into the second coupling groove 2123, and the driving shaft 212 is coupled to the rotating cylinder 214 so as to rotate together with the rotating cylinder 214.

FIG. 20 is a view showing a cross-section based on rotation of the roller 210, and shows a state in which the fixed shaft 211 is omitted. When the driving shaft 212 rotates relative to the fixed shaft 211, the number of turns of the torsion spring changes, and the torsion spring provides the elasticity to restore the original number of turns. In (a) in FIG. 20, in the first state, one end 2131 and the other end 2132 of the coil spring 213 may extend in the radial direction and may have 90° therebetween (may not be limited to 90° and may have a different angle).

When the state of the mobile terminal 100 is switched, the roller 210 rotates. At this time, only the driving shaft 212 and the rotating cylinder 214 rotate, and the fixed shaft 211 does not rotate. Because the number of turns of the coil spring 213 changes, the angle between the bent one end 2131 and the other end of the coil spring 213 changes as shown in (b) in FIG. 20.

When the state of the mobile terminal 100 is switched from the first state to the second state, the driving unit 200 moves the second frame 102 in the first direction, the variable portion of the display unit 151 moves, and the second region moves to the front of the mobile terminal 100. The support bars 1041 of the rolling hinge 104 located on the rear surface of the display unit 151 rotates the rotating cylinder 214 of the roller 210 by being engaged with the teeth 2141 of the roller 210.

The driving shaft 212 coupled to the rotating cylinder 214 rotates together, and the number of turns of the coil spring 213 changes. The coil spring 213 whose number of turns has changed has the elasticity to restore the original number of turns, and provides a force for the roller 210 to rotate in an opposite direction.

However, because the motor of the driving unit 200 restricts the movement of the second frame 102, the roller 210 is not able to rotate in the opposite direction. However, a pulling force may be provided to the display unit 151 such that the third region located at the front of the display unit 151 maintains a maximum flat state.

Conversely, when the state is switched from the second state to the first state, the driving unit 200 moves the second frame 102 in the second direction. When the second frame 102 moves in the first direction, the third region of the

28

display unit 151 moves to the front and thus the slide frame 103 naturally moves in the first direction. However, when the second frame 102 conversely moves in the second direction, the force of the driving unit 200 is not directly transmitted to the slide frame 103, so that the movement of the slide frame 103 in the second direction may be slower than the movement of the second frame 102 in the second direction.

The coil spring 213 of the roller 210 provides a force for the third region of the display unit 151 to move to the rear of the mobile terminal 100 when the state is switched from the second state to the first state. More specifically, when the second frame 102 moves in the second direction, as the force of the driving unit 200 fixing the second frame 102 is removed, the coil spring 213 rotates the driving shaft 212 and the rotating cylinder 214 in the opposite direction so as to move the third region of the display unit 151 in a rearward direction corresponding to a moving speed of the second frame 102.

Thus, because of the roller 210 including the coil spring 213, the movement of the slide frame 103 may be synchronized with the movement of the second frame 102, and the display unit 151 may maintain the flat state.

FIG. 21 is a view showing arrangement of the rollers 210 coupled to the second frame 102 of the present disclosure. A region where the driving unit 200 is disposed may be disposed above the roller 210, and the roller 210 may include a plurality of rollers spaced apart from each other in the third direction.

As described above, the mobile terminal 100 of the present disclosure may adjust the size of the screen as needed, so that both portability and usability may be obtained.

In addition, in the mobile terminal 100 of the present disclosure, because stress is not concentrated at a specific point of the display unit 151, damage to the display unit 151 may be reduced.

In addition, the display unit 151 may be prevented from lifting when the second frame 102 moves, so that the display unit 151 in the flat state may be maintained.

In addition, when the state is switched from the second state to the first state, the roller 210 itself may provide a force to restore the first state to the display unit, so that the display unit may move in synchronization with the movement of the second frame 102.

The above-described embodiments are to be construed in all aspects as illustrative and not restrictive. The scope of the disclosure should be determined by the appended claims and their legal equivalents, not by the above description, and all changes coming within the meaning and equivalency range of the appended claims are intended to be embraced therein.

What is claimed is:

1. A mobile terminal capable of switching a state thereof from a first state to a second state extended in a first direction, the mobile terminal comprising:

a first frame;

a second frame capable of sliding in a first direction or in a second direction opposite to the first direction relative to the first frame and including a roller seating portion at an end of the second frame in the first direction;

a roller coupled to the roller seating portion;

a flexible display unit including a fixed portion coupled to the first frame and a variable portion extending in a rearward direction from the fixed portion while surrounding the roller and having an area located at a front variable based on the sliding of the second frame; and

29

a rolling hinge located on a rear surface of the variable portion and changeable based on a shape of the variable portion,
 wherein the roller includes teeth formed on an outer surface thereof,
 wherein the rolling hinge includes irregularities corresponding to the teeth,
 wherein the roller includes:
 a fixed shaft fixed to the second frame;
 a driving shaft rotatable relative to the fixed shaft;
 a coil spring having one end coupled to the fixed shaft and another end coupled to the driving shaft; and
 a rotating cylinder rotatable together with the driving shaft and having the teeth formed thereon, and
 wherein the roller seating portion includes:
 a fixing hole for receiving a screw fastened through one end of the fixed shaft; and
 a bearing for one end of the driving shaft to be rotatably seated.

2. The mobile terminal of claim 1, wherein the rolling hinge includes a plurality of support bars extending in a third direction perpendicular to the first direction and arranged adjacent to each other in the first direction,
 wherein the teeth of the roller include each protrusion inserted between adjacent two of the plurality of support bars.

3. The mobile terminal of claim 1, wherein the rotating cylinder and the driving shaft rotate when the second frame moves and change the number of turns of the coil spring,

30

wherein the coil spring provides elasticity to restore a reference number of turns to the driving shaft and the rotating cylinder.

4. The mobile terminal of claim 3, wherein the elasticity of the coil spring is provided when restoring the first state from the second state.

5. The mobile terminal of claim 1, wherein at least a portion of the driving shaft rotatably overlaps the fixed shaft inside of the coil spring.

6. The mobile terminal of claim 1, wherein the fixed shaft includes a first coupling groove for receiving the one end of the coil spring,

wherein the driving shaft includes a second coupling groove for receiving the other end of the coil spring.

7. The mobile terminal of claim 6, wherein both of the ends of the coil spring are bent in a radial direction of the coil,

wherein the first coupling groove and the second coupling groove extend through a center of rotation of the driving shaft.

8. The mobile terminal of claim 7, wherein the first coupling groove and the second coupling groove respectively extend to another end of the fixed shaft and another end of the driving shaft.

9. The mobile terminal of claim 1, wherein the roller includes a plurality of rollers spaced apart from each other in a third direction perpendicular to the first direction.

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